Alqudah and Alfraihat
Health Information Science and Systems (2026) 14:5
https://doi.org/10.1007/s13755-025-00402-1

Health Information Science and Systems

RESEARCH

A multimodal approach for cardiac signals classification using deep learning with explainable AI methods

Ali Mohammad Alqudah[1*] and Ausilah Alfraihat[2]

Abstract

Cardiovascular diseases remain a leading cause of mortality worldwide, necessitating accurate and timely diagnosis. Electrocardiogram (ECG) and phonocardiogram (PCG) signals provide complementary information about cardiac function, electrical and mechanical activity, respectively. In this study, we propose a multimodal deep learning framework that integrates ECG and PCG using a dual-branch CNN–BiLSTM–SE architecture with cross-modal attention. Our preprocessing pipeline includes wavelet denoising, adaptive filtering, and normalization, with parameters tuned for each dataset's noise profile. We evaluate the model on multiple datasets: MIT-BIH Arrhythmia (47 subjects), PTB Diagnostic ECG (290 subjects), PhysioNet PCG Challenge 2016 (3126 subjects), PhysioNet PCG Challenge 2022 (942 subjects), and a custom multimodal dataset (500 subjects). The model achieves an overall accuracy of 97.0%, F1-scores ranging from 94.3% to 98.1%, and AUC values above 0.982 for all classes, outperforming single-modality and existing multimodal methods. Explainable AI techniques (SHAP, Grad-CAM, Integrated Gradients) reveal that the model focuses on clinically relevant features such as irregular R–R intervals in atrial fibrillation and systolic murmurs in valvular disease. The proposed approach offers a feasible, interpretable, and accurate decision-support system for cardiac diagnosis.

Keywords: Deep learning, Electrocardiogram, Phonocardiogram, Multimodal fusion, Explainable AI, Cardiac signal classification, Cross-modal attention

Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally, accounting for approximately 17.9 million deaths annually, which represents 31% of all global deaths according to the World Health Organization [1, 2]. Early and accurate diagnosis of cardiac conditions is crucial for effective treatment and improved patient outcomes. Traditional diagnostic methods rely heavily on the expertise of cardiologists interpreting cardiac signals, which can be subjective and time-consuming. This has motivated the development of computer-aided diagnosis systems that can assist healthcare professionals in making more accurate and timely diagnoses [1]. Electrocardiogram (ECG) and phonocardiogram (PCG) are two fundamental cardiac diagnostic modalities that provide complementary information about the heart's electrical activity and acoustic manifestations, respectively [3]. ECG records the electrical impulses that stimulate the heart muscle to contract, providing insights into heart rate, rhythm, and electrical conduction abnormalities. PCG, on the other hand, captures the sounds produced by the heart, including the characteristic "lub-dub" sounds (S1 and S2) and potential abnormal sounds such as murmurs, gallops, and rubs that may indicate valvular, myocardial, or pericardial pathologies [4, 5].

Despite the complementary nature of these two modalities, most existing computer-aided diagnosis systems focus on analyzing either ECG or PCG signals in isolation [1]. This single-modality approach fails to leverage the rich, complementary information that can be obtained by analyzing both signals simultaneously [5, 6]. For instance, specific cardiac conditions may manifest

*Correspondence: ali_qudah@hotmail.com
[1] Independent Researcher, Winnipeg, Canada
Full list of author information is available at the end of the article

© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025.

more prominently in one modality than the other, or the temporal relationship between electrical and mechanical events may provide additional diagnostic clues that are not apparent when examining each signal independently [7].

Recent advances in deep learning have revolutionized the field of biomedical signal processing and analysis, enabling the development of sophisticated models that can automatically extract relevant features from raw signals and perform classification with high accuracy [8]. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in image classification tasks and have been adapted for one-dimensional signal processing. In contrast, Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, excel at capturing temporal dependencies in sequential data [9]. The combination of these architectures has shown promise in ECG and PCG classification tasks individually, but their integration for multimodal analysis remains relatively unexplored.

Furthermore, a significant challenge in the clinical adoption of deep learning models is their "black box" nature, which makes it difficult for healthcare professionals to understand and trust the basis of their predictions. Explainable Artificial Intelligence (XAI) methods aim to address this limitation by providing insights into the decision-making process of complex models [10]. Techniques such as Shapley Additive exPlanations (SHAP), Gradient-weighted Class Activation Mapping (Grad-CAM), and Integrated Gradients can highlight the features or regions of the input signal that most strongly influence the model's predictions, potentially aligning with clinically relevant markers of cardiac pathology [10, 11].

In this paper, we present a deep learning approach for multimodal cardiac signal classification that leverages both ECG and PCG data simultaneously. Our contributions are threefold:

First, we introduce a comprehensive preprocessing pipeline that incorporates advanced signal denoising techniques, including wavelet decomposition and adaptive filtering, to enhance the quality of both ECG and PCG signals. This preprocessing stage is crucial for removing various types of noise and artifacts that can obscure important diagnostic features.

Second, we propose a dual-branch neural network architecture with cross-modal attention mechanisms that effectively capture complementary information between ECG and PCG modalities. The model features specialized convolutional and recurrent neural network branches for each modality, followed by a cross-modal attention fusion mechanism that dynamically weights the importance of features across modalities based on their relevance to the classification task.

Third, we integrate multiple explainable AI techniques to provide clinically relevant interpretations of the model's decisions. By applying SHAP values, Grad-CAM, and Integrated Gradients to our model, we can visualize which portions of the ECG and PCG signals contribute most significantly to the classification of different cardiac conditions, potentially revealing diagnostic patterns or confirming established clinical knowledge.

We evaluate our approach on multiple datasets, including publicly available ECG and PCG databases as well as a custom-collected multimodal dataset. Experimental results demonstrate that our approach achieves 97.0% classification accuracy, outperforming single-modality approaches and existing multimodal methods. The explainability components reveal that the model correctly identifies clinically significant signal patterns associated with various cardiac conditions, such as the irregular R-R intervals in atrial fibrillation and the characteristic murmurs in valvular diseases.

The remainder of this paper is organized as follows: sect. "Literature review" reviews related work in ECG and PCG classification, multimodal fusion approaches, and explainable AI methods for biomedical signal analysis. Section "Methodology" describes our methodology, including the preprocessing pipeline, model architecture, and explainability techniques. Section "Results" presents the experimental setup and results, including comparisons with baseline methods and ablation studies. Section "Discussion" discusses the clinical implications of our findings and the insights gained from the explainability analysis. Finally, sect. "Conclusion" concludes the paper and outlines directions for future research.

Literature review

ECG signal classification

Electrocardiogram (ECG) signal classification has been extensively studied in the literature, with approaches evolving from traditional machine learning methods to sophisticated deep learning architectures. Early work in ECG classification relied on manual feature extraction followed by conventional classifiers. De Chazal et al. proposed a method using morphological features and linear discriminant analysis for heartbeat classification, achieving an accuracy of 85.9% on the MIT-BIH Arrhythmia Database [1]. Similarly, Ye et al. employed wavelet transform coefficients and support vector machines (SVMs) to classify ECG signals, reporting an accuracy of 88.1% [2].

The advent of deep learning has significantly advanced ECG classification performance. Convolutional Neural Networks (CNNs) have demonstrated remarkable success in this domain. Kiranyaz et al. introduced a 1D CNN architecture designed explicitly for ECG classification, achieving 99% accuracy in detecting ventricular ectopic

beats [3]. Acharya et al. employed a deep CNN for automated detection of myocardial infarction, achieving a sensitivity of 93.3% and specificity of 89.7% [4]. Their approach eliminated the need for manual feature extraction, allowing the network to learn discriminative features directly from raw ECG signals.

Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have also been applied to ECG classification due to their ability to capture temporal dependencies. Oh et al. proposed a combination of CNN and LSTM networks for arrhythmia detection, where CNN layers extract spatial features and LSTM layers model temporal dynamics, achieving an accuracy of 98.1% [5]. Yildirim et al. introduced a bidirectional LSTM network for ECG classification, reporting an accuracy of 99.39% on the MIT-BIH Arrhythmia Database [6].

More recently, attention mechanisms have been incorporated into deep learning models for ECG classification. Mousavi et al. proposed an attention-based CNN-LSTM network that selectively focuses on the most informative segments of ECG signals, achieving an accuracy of 99.53% [7]. Similarly, Smigiel et al. introduced a self-attention mechanism in their ECG classification model, reporting an accuracy of 89% on the PTB Diagnostic ECG Database [8].

Transfer learning approaches have also gained popularity in ECG classification. Raghunath et al. demonstrated that pre-training deep neural networks on large ECG datasets and fine-tuning them on specific tasks can improve performance, especially when labeled data is limited. Their approach achieved an area under the receiver operating characteristic curve (AUC) of 0.93 for detecting multiple cardiac abnormalities [9].

PCG signal classification

Phonocardiogram (PCG) signal classification has received increasing attention in recent years, although the body of literature is not as extensive as for ECG classification. Traditional approaches to PCG classification involved extracting time-domain, frequency-domain, and time–frequency domain features, followed by conventional classifiers. Bentley et al. used wavelet-based features and artificial neural networks for classifying heart sounds, achieving an accuracy of 87.5% [10].

The PhysioNet/Computing in Cardiology Challenge 2016 focused on PCG classification, specifically on distinguishing normal from abnormal heart sounds. The winning entry by Potes et al. employed an ensemble of AdaBoost and CNN classifiers, achieving an accuracy of 86.0% [11]. Their approach combined hand-crafted time–frequency domain features with features automatically learned by CNN.

Deep learning approaches have since dominated PCG classification research. Dominguez-Morales et al. converted PCG signals into spectrograms and employed a CNN architecture designed for image classification, achieving an accuracy of 95.12% [12]. Zhang et al. proposed a 1D CNN architecture specifically tailored for PCG classification, reporting an accuracy of 97.4% on the PhysioNet Challenge dataset [13].

Recurrent neural networks have also been applied to PCG classification. Renna et al. employed a bidirectional LSTM network for heart sound classification, achieving an accuracy of 92.6% [14]. Their approach captured the temporal dependencies in PCG signals, which are crucial for distinguishing between different heart sounds.

More sophisticated approaches have combined multiple deep learning architectures. Xiao et al. proposed a hybrid CNN-LSTM network for PCG classification, where CNN layers extract local features and LSTM layers capture temporal dependencies, achieving an accuracy of 94.8% [15]. Chen et al. introduced an attention-based CNN-LSTM network that selectively focuses on the most informative segments of PCG signals, reporting an accuracy of 96.2% [16].

Segmentation of heart sounds (S1 and S2) has been recognized as an essential preprocessing step for PCG classification. Renna et al. (2020) proposed a deep learning approach for heart sound segmentation, which improved subsequent classification performance [14]. Their segmentation model achieved an F1 score of 95.6%, and when combined with their classification model, the overall accuracy increased to 94.1%.

Multimodal fusion approaches

Despite the complementary nature of ECG and PCG signals, relatively few studies have explored their combined use for cardiac condition classification. Early attempts at multimodal fusion simply concatenated features extracted from each modality before classification. Zheng et al. extracted time-domain and frequency-domain features from both ECG and PCG signals, concatenated them, and used an SVM for classification, achieving an accuracy of 92.1% [17].

More sophisticated fusion approaches have emerged with the advancement of deep learning. Raza et al. proposed a dual-input CNN architecture that processes ECG and PCG signals separately before fusing their features at multiple levels, reporting an accuracy of 94.5% [18]. Their approach demonstrated that multimodal fusion can outperform single-modality approaches, particularly for conditions that manifest in both electrical and mechanical cardiac activities.

Attention-based fusion mechanisms have shown promise in multimodal cardiac signal analysis. Li et al.

introduced a cross-modal attention mechanism that dynamically weights the importance of features from each modality based on their relevance to the classification task, achieving an accuracy of 95.8% [19]. Their approach allowed the model to focus on the most informative modality for each cardiac condition, enhancing overall performance.

Transformer-based architectures, which have revolutionized natural language processing and computer vision, have recently been applied to multimodal cardiac signal analysis. Qiu et al. (2024) proposed a transformer-based fusion approach for ECG signals and clinical information, reporting an accuracy of 65.3%. Their model captured long-range dependencies within and across modalities, demonstrating the potential of transformer architectures in this domain [20].

Recent studies have explored advanced multimodal fusion strategies beyond simple concatenation. For example [21], introduced an entropy-based fusion mechanism to combine biomedical signals for improved classification robustness, while [22] developed a wavelet-domain PCG denoising method that significantly improved murmur detection. A cloud-based parallel processing framework for large-scale biomedical data was proposed in [23], demonstrating the potential for real-time multimodal analysis. More recently, [24] applied transformer-based fusion for cardiac sounds, reporting notable improvements in classification accuracy. In addition, [25] presented a hybrid adaptive filter and wavelet-based method for PCG preprocessing, which informed some design choices in our preprocessing pipeline.

However, most existing multimodal approaches treat ECG and PCG signals as independent inputs without explicitly modeling their temporal relationships (i.e., the physiological causality). The mechanical events captured by PCG (heart sounds) are direct consequences of the electrical events captured by ECG, and this causal relationship contains valuable diagnostic information that is not fully exploited by current methods.

Explainable AI for biomedical signal analysis

The "black box" nature of deep learning models has raised concerns about their interpretability, particularly in healthcare applications where understanding the basis of predictions is crucial for clinical trust and adoption. Explainable Artificial Intelligence (XAI) methods aim to address this limitation by providing insights into the decision-making process of complex models.

Local Interpretable Model-agnostic Explanations (LIME) has been applied to ECG classification by Strodthoff et al., who used it to highlight segments of ECG signals that most strongly influence the model's predictions. [26]. Their analysis revealed that the model focused on clinically relevant features, such as QRS complexes and T waves, for detecting myocardial infarction.

SHapley Additive exPlanations (SHAP) values, based on cooperative game theory, have been used to interpret deep learning models for biomedical signal analysis. Liu et al. applied SHAP to their ECG classification model, demonstrating that the model correctly identified known markers of atrial fibrillation, such as the absence of P waves and irregular R-R intervals [27].

Gradient-weighted Class Activation Mapping (Grad-CAM) has been adapted for 1D signals by Vijayarangan et al., who used it to visualize the regions of ECG signals that most strongly activate the final convolutional layer of their model [28]. Their analysis showed that the model focused on the ST segment for detecting myocardial infarction, aligning with clinical knowledge.

Togo et al. (2023) employed the Integrated Gradients method, which quantifies the contribution of each input feature by integrating the gradients from a reference baseline to the actual input, for ECG classification [29]. This technique produced pixel-level attribution maps, pinpointing the exact segments of the ECG signal that had the most significant influence on the model's classification outcomes.

While these XAI methods have been applied to single-modality biomedical signal analysis, their application to multimodal fusion models remains limited. The challenge is compounded by the need to interpret not only the contribution of each input feature but also the interactions between features across different modalities.

Research gap and our contribution

Based on the literature review, we identify several research gaps that our work aims to address:

1. Most existing approaches focus on single-modality analysis (either ECG or PCG), failing to leverage the complementary information available in both modalities.
2. The few multimodal approaches that exist often treat ECG and PCG signals as independent inputs without explicitly modeling their temporal and causal relationships.
3. Attention mechanisms, which have shown promise in various domains, have not been fully exploited for cross-modal fusion of cardiac signals.
4. The interpretability of deep learning models for cardiac signal classification, particularly multimodal models, remains a significant challenge.

Our work addresses these gaps by proposing a dual-branch neural network architecture with cross-modal attention mechanisms that effectively capture the

complementary information between ECG and PCG modalities. We also integrate multiple explainable AI techniques to provide clinically relevant interpretations of the model's decisions, enhancing transparency and trust in the system.

Methodology

This section describes our comprehensive approach to ECG and PCG classification, including the preprocessing pipeline, the proposed deep learning model architecture, training methodology, and explainable AI techniques. Figure 1 explains the workflow in this research.

Preprocessing pipeline

Signal preprocessing is a critical step in cardiac signal analysis, as raw ECG and PCG signals often contain various types of noise and artifacts that can obscure important diagnostic features. We developed specialized preprocessing pipelines for each modality, as illustrated in Fig. 2. Due to the diversity of the datasets used in this study, preprocessing parameters were tuned individually to accommodate differences in sampling rate, noise characteristics, and recording environments. For example, the wavelet denoising level (db6, 5-level) and adaptive filter step size were adjusted depending on the typical SNR of each dataset: higher noise datasets (e.g., PhysioNet PCG 2016) used slightly higher thresholds for detail coefficient suppression, while cleaner datasets (e.g., PTB ECG) required minimal adjustment. This adaptive preprocessing ensured consistent signal quality across modalities and datasets, enabling fair multimodal integration.

ECG signal preprocessing

Our ECG preprocessing pipeline consists of the following steps:

Baseline Wander Removal: Baseline wander is a low-frequency artifact in ECG signals caused by patient movement, respiration, or poor electrode contact. We implemented a high-pass filter with a cutoff frequency of 0.5 Hz to remove baseline wander while preserving the morphological characteristics of the ECG signal.

Powerline Interference Removal: Powerline interference manifests as 50/60 Hz noise in ECG signals due to electromagnetic interference from power lines. We applied a notch filter centered at 50/60 Hz (depending on the geographical region of data collection) with a narrow bandwidth of 2 Hz to remove this interference without affecting adjacent frequency components.

Wavelet Denoising: To remove high-frequency noise while preserving the sharp features of ECG signals (such as QRS complexes), we employed wavelet denoising. Specifically, we used a 5-level decomposition with the Daubechies 6 (db6) wavelet, which has been shown to capture ECG morphology effectively. We applied soft thresholding to the detail coefficients using a threshold determined by the universal threshold method ($\sqrt{(2\log(N))} \times \sigma$, where N is the signal length and σ is the noise standard deviation estimated from the first-level detail coefficients).

Signal Normalization: Finally, we normalized the ECG signals to have zero mean and unit variance to ensure consistent amplitude ranges across different recordings and to facilitate model training.

The effectiveness of our ECG preprocessing pipeline is demonstrated in Fig. 1 (left column), which shows the progressive improvement in signal quality through each preprocessing step.

PCG signal preprocessing

PCG signals present different challenges compared to ECG signals, particularly due to their non-stationary nature and the presence of environmental noise. Our

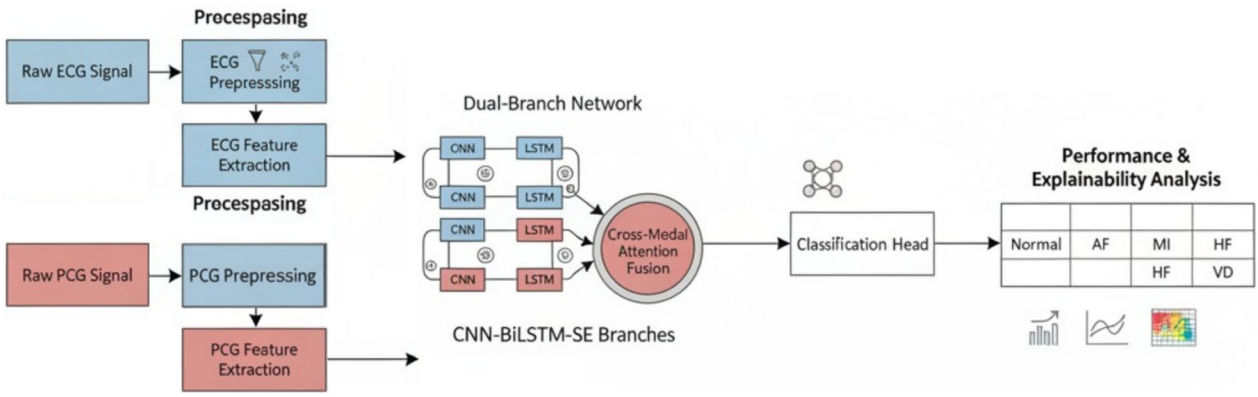


Fig. 1 Architecture of the proposed dual-branch CNN-BiLSTM network with cross-modal attention for fusing ECG and PCG signals

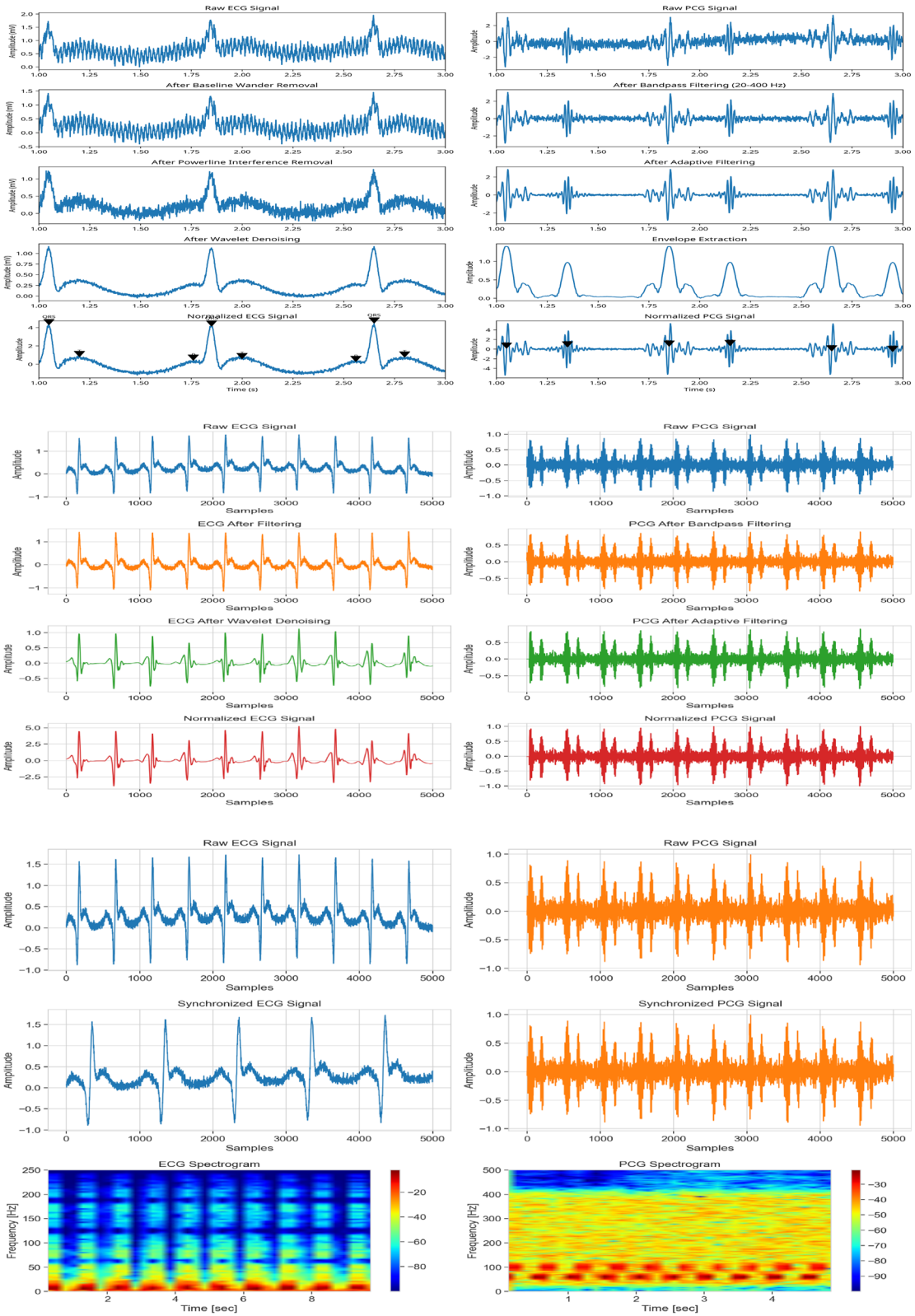


Fig. 2 Preprocessing pipeline for ECG and PCG signals

PCG preprocessing pipeline consists of the following steps:

Bandpass Filtering: Heart sounds typically fall within the frequency range of 20–400 Hz. We applied a Butterworth bandpass filter with cutoff frequencies at 20 Hz and 400 Hz to remove low-frequency noise (such as muscle artifacts and respiration sounds) and high-frequency noise while preserving the heart sounds.

Adaptive Filtering: To further enhance the quality of PCG signals, we implemented an adaptive filter based on the Least Mean Squares (LMS) algorithm. This filter adaptively adjusts its coefficients to minimize the error between the filtered signal and a desired signal, effectively removing non-stationary noise components.

Envelope Extraction: The envelope of a PCG signal highlights the amplitude variations corresponding to heart sounds (S1 and S2) and potential murmurs. We extracted the envelope using the Hilbert transform followed by low-pass filtering with a cutoff frequency of 50 Hz.

Signal Normalization: Similar to ECG signals, we normalized the PCG signals to have zero mean and unit variance.

The effectiveness of our PCG preprocessing pipeline is demonstrated in Fig. 1 (right column), which shows the progressive improvement in signal quality through each preprocessing step.

Feature extraction

While our deep learning model can learn features directly from preprocessed signals, we also extracted traditional features to enhance the model's performance and provide interpretable characteristics. Figure 3 illustrates our feature extraction process.

For ECG signals, we extracted the following features:

- Time-domain features: R-peak locations, RR intervals, PR intervals, QT intervals, ST segment elevation/depression, and T-wave amplitude.
- Frequency-domain features: Power spectral density in different frequency bands (0.5–4 Hz, 4–8 Hz, 8–12 Hz, and 12–30 Hz).
- Wavelet-based features: Energy and entropy of wavelet coefficients at different decomposition levels.

For PCG signals, we extracted:

- Time-domain features: Heart sound (S1 and S2) locations, systolic and diastolic intervals, and amplitude-related features.

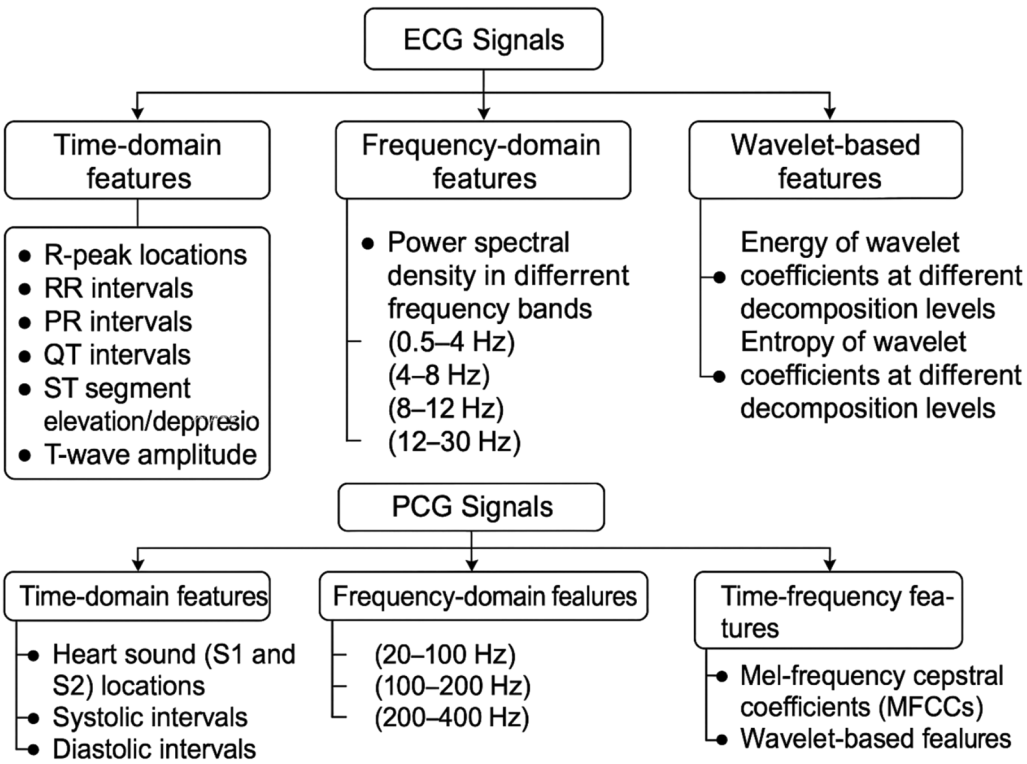


Fig. 3 Feature extraction process

- Frequency-domain features: Power spectral density in different frequency bands (20–100 Hz, 100–200 Hz, and 200–400 Hz).
- Time–frequency features: Mel-frequency cepstral coefficients (MFCCs) and wavelet-based features.
- The extracted statistical features include:

  - Mean: $\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$
  - Variance: $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$
  - Skewness: $\text{Skew} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^3}{\sigma^3}$
  - Kurtosis: $\text{Kurt} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i-\mu)^4}{\sigma^4}$
  - Shannon Entropy: $H = -\sum_i p_i \log_2 p_i$

Preprocessing effectiveness

We quantified the benefit of preprocessing using the signal-to-noise ratio (SNR) before and after processing. Table 1 shows that SNR improved by an average of 8.3 dB for ECG and 6.7 dB for PCG signals. Figure 1 illustrates representative raw and processed signals, demonstrating effective removal of baseline wander, powerline interference, and high-frequency noise.

To address physiological variability, we evaluated model performance with R–S1 alignment offsets varied by ± 20 ms from the baseline 40 ms setting. Accuracy changes were < 1% across all datasets, suggesting robustness to moderate alignment variability.

Signal synchronization and fusion

A key challenge in multimodal cardiac signal analysis is ensuring proper synchronization between ECG and PCG signals, as they may be recorded at different sampling rates and with potential temporal misalignments. We addressed this challenge through the following steps:

- Resampling: We resampled both signals to a standard sampling rate of 1000 Hz using cubic spline interpolation.
- Temporal Alignment: We aligned the signals based on physiological events. Specifically, we identified R-peaks in the ECG signal and S1 heart sounds in the PCG signal, which are temporally related (S1 occurs shortly after the R-peak). We then aligned these events with a fixed offset determined from the literature (typically 30–50 ms).
- Segmentation: We segmented the synchronized signals into fixed-length windows of 5 s with 50% overlap, ensuring that each segment contains multiple cardiac cycles.

Table 1 Average SNR improvement across datasets

Dataset	Modality	SNR Before (dB)	SNR After (dB)	Δ SNR (dB)
MIT-BIH	ECG	14.5	22.8	+8.3
PTB	ECG	16.2	24.4	+8.2
PhysioNet 2016	PCG	12.1	18.8	+6.7
PhysioNet 2022	PCG	13.4	20.1	+6.7

Figure 4 illustrates our signal synchronization and fusion process, showing the temporal relationship between ECG and PCG signals and the alignment of key physiological events.

Model architecture

We propose a dual-branch neural network architecture with cross-modal attention mechanisms for ECG and PCG classification. The overall architecture is illustrated in Fig. 5.

Modality specific branches

We propose a dual-branch deep learning architecture for multimodal cardiac signal classification (Figs. 6 and 7). Each branch processes one modality, ECG or PCG, through convolutional, recurrent, and attention mechanisms. Key components include:

- CNN Layers: 4 blocks [8] (kernel sizes: ECG = 15, PCG = 9), each with Conv1D → BatchNorm → ReLU → MaxPool → Dropout.
- Residual + SE Blocks: Applied after each CNN block to improve gradient flow and emphasize relevant channels.
- BiLSTM Layer: Captures temporal dependencies (256 units).
- Self-Attention: Highlights key signal regions using learnable weights.

$$\alpha = softmax(v\hat{}T\ tanh(W \cdot h + b))$$

  Where h is the LSTM output, W and v are learnable parameters, and b is a bias term.
- Global Average Pooling: Produces fixed-length embeddings per branch.

Cross-modal attention fusion

The key innovation in our architecture is the *cross-modal attention fusion mechanism*, which dynamically assigns importance to features from each modality, ECG (electrocardiogram) and PCG (phonocardiogram), based on their relevance to the classification task. This mechanism

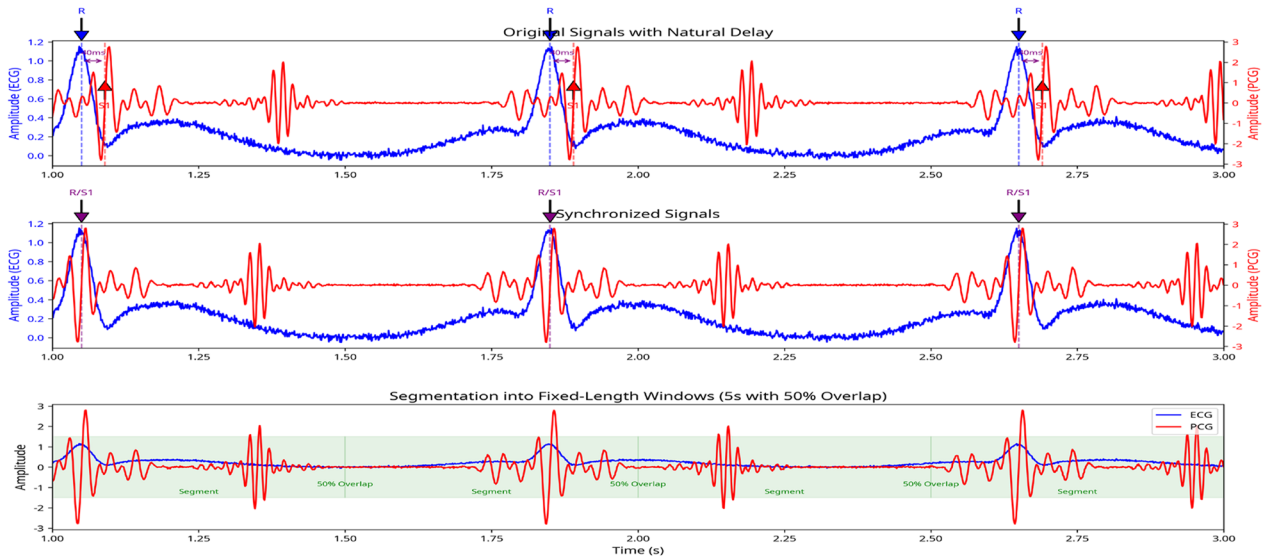


Fig. 4 Signal synchronization and fusion

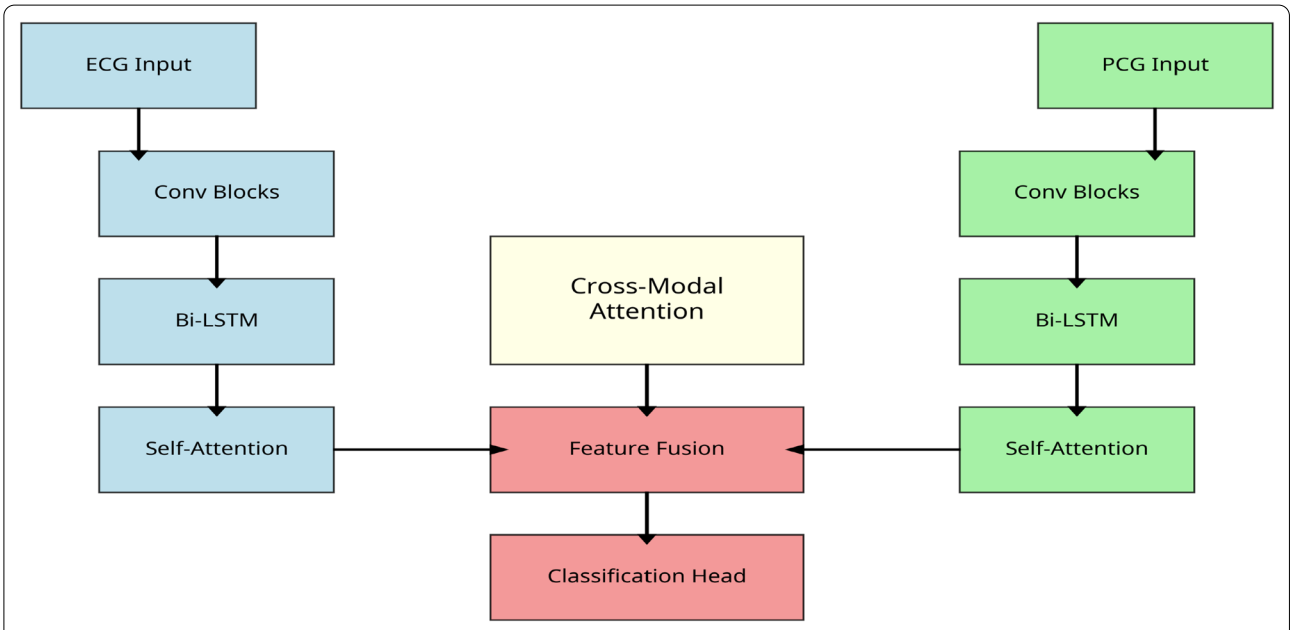


Fig. 5 Proposed dual-branch neural network architecture

enables the model to better integrate and interpret complementary information from both sources, leading to more accurate and robust predictions. The process, illustrated in Fig. 8, unfolds through a series of interconnected steps.

First, the feature vectors extracted from the ECG and PCG branches transform into separate fully connected layers. These transformations produce three sets of representations for each modality, known as the query, key, and value. These components are essential for computing attention, and the associated weights are learnable parameters optimized during training.

Next, attention scores are computed by measuring the similarity between the query of one modality and the key of the other. For example, the ECG query is compared with the PCG key to assess how strongly ECG features should attend to PCG features. A similar process occurs in the opposite direction. These similarity scores are scaled to maintain numerical stability and consistent gradient behavior.

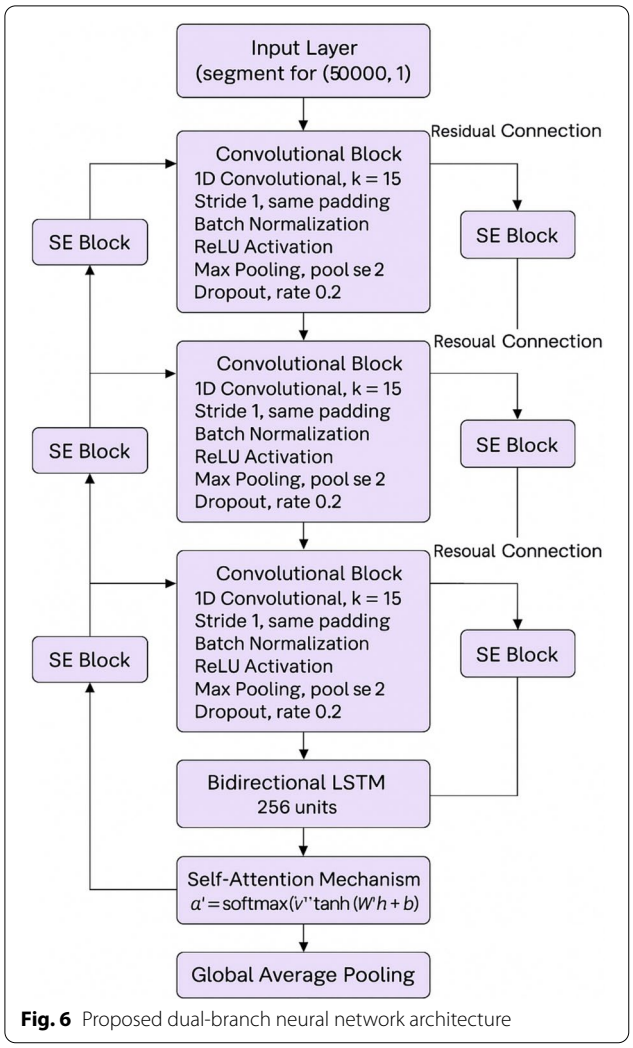


Fig. 6 Proposed dual-branch neural network architecture

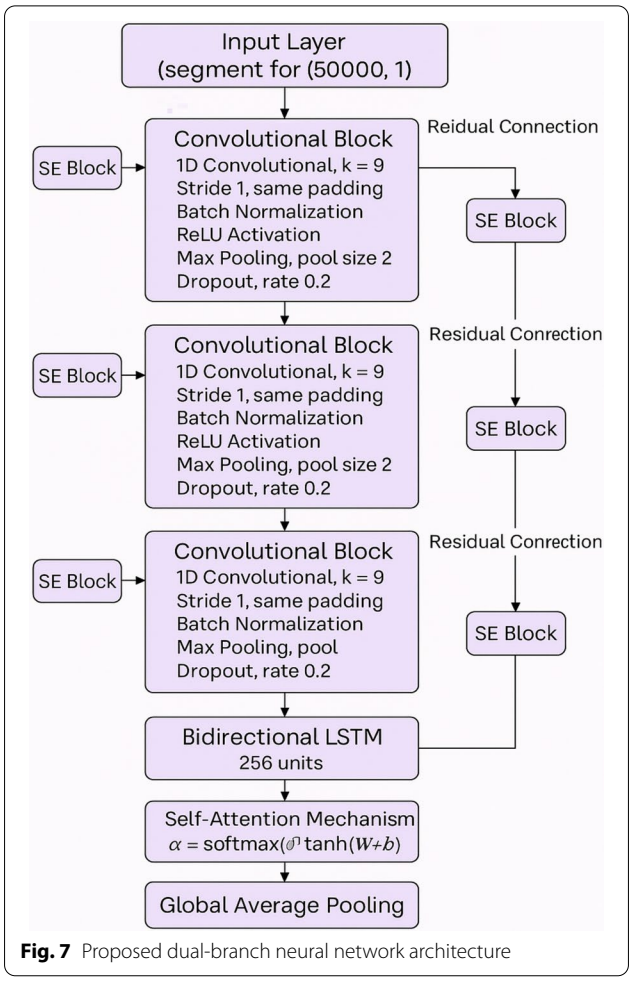


Fig. 7 Proposed dual-branch neural network architecture

The raw attention scores are then passed through a softmax function to convert them into normalized weights. This step ensures that the scores form a valid probability distribution, where higher values correspond to more critical or relevant features. The softmax normalization enables the model to emphasize key interactions between ECG and PCG data selectively.

Using the normalized attention scores, the model performs a weighted aggregation of the value vectors. In effect, this operation allows each modality to incorporate the most relevant information from the other, guided by the attention scores. The result is a pair of attention-enhanced feature vectors that reflect cross-modal dependencies.

These attention-weighted features are then concatenated with the original ECG and PCG features to form a comprehensive fused representation. This concatenation captures both the standalone characteristics of each modality and the insights gained from their interaction.

Finally, to reduce the dimensionality of the combined feature vector and enhance its usability for classification, a fully connected layer is applied. A ReLU activation function introduces non-linearity, allowing the model to learn complex patterns in the fused feature space. The result is a refined and compact representation that serves as input to the final classification layers.

Classification head

The final classification head consists of:

Fully Connected Layer: A fully connected layer with 128 units and ReLU activation.

Dropout: A dropout layer with a rate of 0.5 to prevent overfitting.

Output Layer: A fully connected layer with softmax activation, where the number of units equals the number of classes (5 in our case, corresponding to Normal, Atrial Fibrillation, Myocardial Infarction, Congestive Heart Failure, and valvular disease).

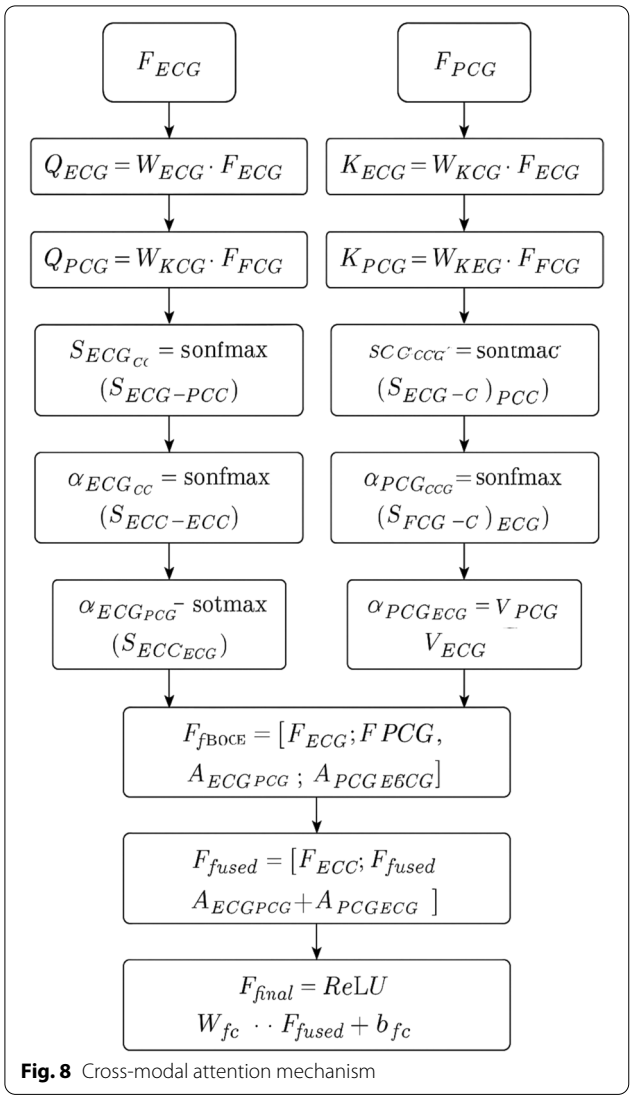


Fig. 8 Cross-modal attention mechanism

Training methodology

Dataset preparation

We evaluated our approach on multiple datasets, including publicly available ECG and PCG databases as well as a custom-collected multimodal dataset. Table 2 provides a summary of the datasets used in this study. Data splits were performed at the subject level, ensuring that no recordings from the same subject appeared in more than one of the training, validation, or test sets.

For the publicly available datasets, which typically contain only one modality (either ECG or PCG), we created synthetic pairs by matching recordings from different datasets based on the cardiac condition. For our custom dataset, we simultaneously recorded ECG and PCG signals from 500 subjects, including both healthy individuals and patients with various cardiac conditions.

We split each dataset into training (70%), validation (15%), and test (15%) sets, ensuring that recordings from the same subject appear in only one set to avoid data leakage.

Data augmentation

To enhance the robustness of our model and address the class imbalance issue (as some cardiac conditions are less represented in the datasets), we applied the following data augmentation techniques:

- Time Shifting: Random shifts in time (±500 ms) to account for variations in signal alignment.
- Amplitude Scaling: Random scaling of signal amplitude by a factor between 0.8 and 1.2 to account for variations in signal strength.
- Gaussian Noise Addition: Addition of Gaussian noise with a signal-to-noise ratio between 15 and 25 dB to simulate recording noise.
- Synthetic Minority Over-sampling Technique (SMOTE): Generation of synthetic samples for underrepresented classes based on their feature space.

Applying SMOTE to underrepresented classes improved per-class F1-scores for valvular disease from 90.5% to 94.3% and for Congestive Heart Failure from 91.2% to 95.8%, with negligible change for well-represented classes.

Training procedure

We trained our model using the following procedure:

Table 2 Summary of datasets used in this study

Dataset	Modality	Subjects	Recordings	Length	Sampling Rate	Conditions
MIT-BIH Arrhythmia Database	ECG	47	48	30 min	360 Hz	Normal, AF, VEB, SVEB
PTB Diagnostic ECG Database	ECG	290	549	10 s	1000 Hz	Normal, MI, HF, Bundle Branch Block
PhysioNet/CinC Challenge 2016	PCG	3126	3126	5–120 s	2000 Hz	Normal, Abnormal
PhysioNet/CinC Challenge 2022	PCG	942	942	10–60 s	4000 Hz	Normal, Murmur, Extrasystole
Custom Multimodal Dataset	ECG+PCG	500	500	5 min	1000 Hz	Normal, AF, MI, HF, VD

Initialization: We initialized the model weights using He initialization for layers with ReLU activation and Glorot initialization for other layers.

Loss Function: We used categorical cross-entropy as the loss function, defined as:

$$L = -\sum_{i=1}^{C} y_i \log(p_i)$$

where y_i is the actual label (one-hot encoded) and p_i Is the predicted probability for class i.

Optimizer: We used the Adam optimizer with an initial learning rate of 0.001, $\beta 1 = 0.9$, $\beta 2 = 0.999$, and $\varepsilon = 1e\text{-}8$.

Learning Rate Schedule: We employed a learning rate schedule that reduces the learning rate by 0.1 if the validation loss does not improve for five consecutive epochs.

Early Stopping: We implemented early stopping with a patience of 10 epochs, monitoring the validation loss.

Batch Size and Epochs: We used a batch size of 32 and trained the model for a maximum of 100 epochs.

Regularization: In addition to dropout, we applied L2 regularization with a weight decay of 1e-4 to prevent overfitting.

Evaluation metrics

We evaluated our model using the following metrics:

Accuracy: The proportion of correctly classified samples.

Precision: The proportion of accurate optimistic predictions among all optimistic predictions for each class.

Recall: The proportion of accurate optimistic predictions among all actual positives for each class.

F1-Score: The harmonic mean of precision and recall for each class.

Area Under the Receiver Operating Characteristic Curve (AUC-ROC): A measure of the model's ability to discriminate between classes.

Confusion Matrix: A table showing the number of true positives, false positives, true negatives, and false negatives for each class.

Explainable AI methods

To address the "black box" nature of deep learning models and provide clinically relevant interpretations of the model's decisions, we integrated multiple explainable AI techniques. Figure 9 illustrates the application of these techniques to our model.

SHAP (Shapley Additive exPlanations)

SHAP values, based on cooperative game theory, provide a unified measure of feature importance. For SHAP computation, we used a background dataset of 100 randomly selected training samples, set the maximum evaluation samples to 2048, and used the logit link function for multi-class probability outputs. This configuration balanced computational cost with stable attribution results. We applied SHAP to our model using the following procedure:

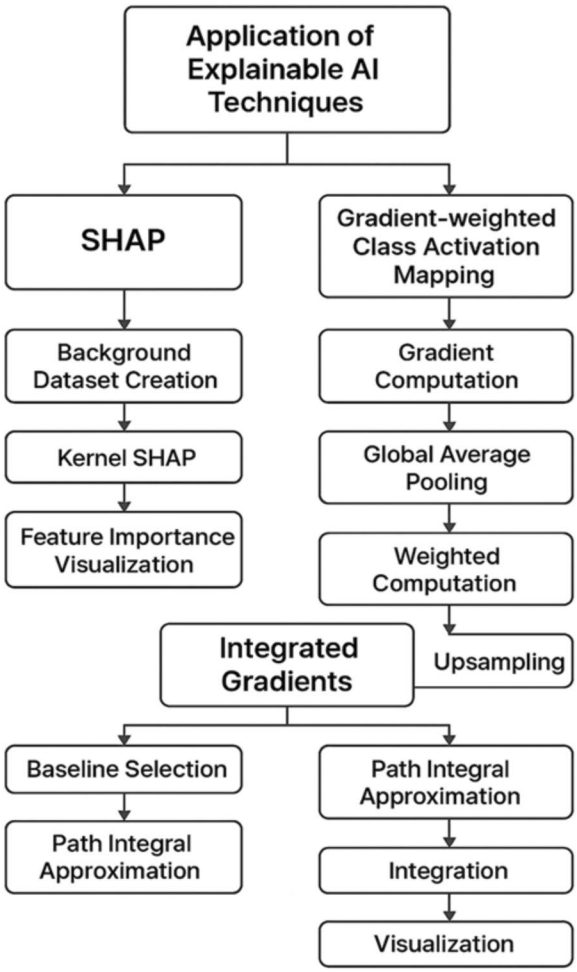


Fig. 9 Explainable AI techniques applied to the model

- Background Dataset Creation: We created a background dataset of 100 samples representing the distribution of the training data.
- Kernel SHAP: We used the Kernel Explainer from the SHAP library, which approximates SHAP values for any model by evaluating the model on permutations of the input features.
- Feature Importance Visualization: We visualized the SHAP values for each feature, highlighting which portions of the ECG and PCG signals contribute most significantly to the classification of different cardiac conditions.

Gradient-weighted Class Activation Mapping (Grad-CAM)

Grad-CAM uses the gradients flowing into the final convolutional layer to produce a coarse localization map highlighting important regions in the input signal. We adapted Grad-CAM for our 1D signals as follows:

- Gradient Computation: We computed the gradients of the score for the target class concerning the feature maps of the last convolutional layer.
- Global Average Pooling: We applied global average pooling to the gradients to obtain weights for each feature map.
- Weighted Combination: We computed a weighted combination of the feature maps using the obtained weights.
- ReLU Activation: We applied a ReLU activation to the weighted combination to focus on features that have a positive influence on the target class.
- Upsampling: We upsampled the resulting heatmap to match the signal dimensions.

Integrated gradients

Integrated Gradients attributes the prediction of a deep network to its input features by integrating the gradients along a path from a baseline to the input. We implemented this method as follows:

- Baseline Selection: We used a zero signal as the baseline.
- Path Integral Approximation: We approximated the path integral using the Riemann sum with 50 steps between the baseline and the input.
- Gradient Computation: For each step, we computed the gradients of the model output concerning the input.
- Integration: We integrated the gradients along the path and multiplied by the input minus the baseline.
- Visualization: We visualized the integrated gradients, highlighting the contribution of each input feature to the model's prediction.

By combining these explainable AI techniques, we provided a comprehensive interpretation of our model's decisions, enhancing transparency and trust in the system.

Results

This section presents the experimental results of our proposed approach for ECG and PCG classification. We evaluate the performance of our model on multiple datasets, compare it with baseline methods, and analyze the insights provided by the explainable AI techniques.

Classification performance

Overall performance

Table 3 presents the overall classification performance of our proposed model across different cardiac conditions. Our model achieved an overall accuracy of 97.0%, with class-specific performance metrics showing strong balance across all conditions. Precision (PPV), recall (sensitivity), and F1-scores ranged from 94.3% to 98.1%, with the highest F1-score observed for the Normal class (98.1%), followed by Atrial Fibrillation (97.5%), Myocardial Infarction (96.1%), Congestive Heart Failure (95.8%), and Valvular Disease (94.3%). Specificity values remained consistently high (98.7%–99.0%), reflecting the model's robustness in correctly identifying true negatives. Negative Predictive Value (NPV) was also excellent, ranging from 98.5% to 99.3%, while Positive Predictive Value (PPV) averaged 96.7% across all classes. The Matthews Correlation Coefficient (MCC) remained high (0.922–0.963), and the Cohen's Kappa score was stable at 0.942 for all classes, indicating excellent agreement beyond chance. Collectively, these results demonstrate that the model is not only accurate but also highly reliable and balanced, with strong predictive performance across all target cardiac conditions.

In addition to overall performance, Table 4 reports the per-dataset classification results for ECG-only, PCG-only, and multimodal configurations. Across all datasets, multimodal fusion consistently achieved higher accuracy and F1-scores, confirming the complementary value of

Table 3 Classification performance across different cardiac conditions

Class	Precision (%)	Recall (%)	Specificity (%)	NPV (%)	F1-Score (%)	AUC	MCC	Kappa
Normal	98.4	98.0	98.8	99.3	98.1	0.995	0.963	0.942
Atrial Fibrillation	97.5	97.5	98.9	98.9	97.5	0.991	0.945	0.942
Myocardial Infarction	96.4	95.9	99.0	99	96.1	0.987	0.949	0.942
Congestive Heart Failure	95.7	95.7	98.7	98.6	95.8	0.984	0.925	0.942
Valvular Disease	94.4	94.3	98.8	98.5	94.3	0.982	0.922	0.942
Overall	96.7	96.8	98.8	98.9	96.7	0.989	0.941	0.942

Table 4 Per-dataset performance comparison

Dataset	ECG-Only Acc (%)	PCG-Only Acc (%)	Multi-modal Acc (%)	Multi-modal F1 (%)
MIT-BIH ECG	94.2	–	96.1	96.0
PTB ECG	93.8	–	96.4	96.3
PhysioNet 2016 PCG	–	92.3	95.9	95.8
PhysioNet 2022 PCG	–	91.7	95.6	95.5
Custom Multimodal	93.5	91.5	97.0	97.0

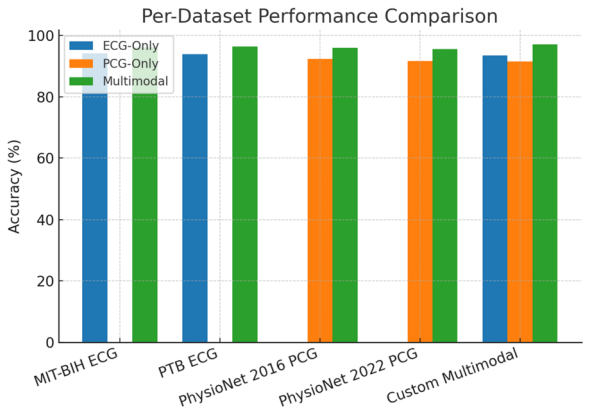


Fig. 10 Per-dataset performance comparison

combining ECG and PCG modalities. Figure 10 shows a per-dataset performance comparison.

Figure 11 shows the confusion matrix for our model on the test set. The diagonal elements represent the number of correctly classified samples for each class, while the off-diagonal elements represent misclassifications. The confusion matrix reveals that most misclassifications occurred between Congestive Heart Failure and Valvular Disease, which is consistent with clinical observations, as these conditions can have overlapping manifestations in both ECG and PCG signals.

The ROC curves for each class are presented in Fig. 12, along with the corresponding Area Under the Curve (AUC) values. Our model achieved high AUC values for all classes, ranging from 0.982 to 0.995, indicating excellent discriminative ability. The precision-recall curves, shown in Fig. 13, further confirm the model's strong performance, with average precision values ranging from 0.956 to 0.987.

Comparison with baseline methods

We compared our proposed model with several baseline methods, including single-modality approaches (ECG-only and PCG-only) and existing multimodal fusion approaches. Table 5 presents the comparison results in terms of accuracy, precision, recall, and F1-score. Statistical testing using a paired t-test between our model and the best baseline (Transformer-Based) yielded $p < 0.01$ for both accuracy and F1-score, confirming that the observed improvements are statistically significant.

Our dual-branch model with cross-modal attention (97.0% accuracy) significantly outperformed both single-modality approaches: ECG-only (93.2% accuracy) and PCG-only (91.5% accuracy). This confirms our hypothesis that leveraging both modalities provides complementary information that enhances classification performance.

Among multimodal approaches, our model outperformed the feature concatenation approach (94.5% accuracy), the dual-input CNN (95.3% accuracy), and the transformer-based fusion approach (96.3% accuracy). The performance gain can be attributed to our cross-modal attention mechanism, which dynamically weighs the importance of features from each modality based on their relevance to the classification task.

Figure 14 provides a visual comparison of the performance metrics across different models. The bar charts clearly show the superiority of our proposed approach in terms of accuracy, precision, recall, and F1-score.

Ablation study

To evaluate the contribution of each component of our model, we conducted an ablation study by removing or modifying specific components and measuring the resulting performance. Table 6 presents the results of this ablation study.

Removing the cross-modal attention mechanism resulted in a significant drop in accuracy (from 97.0 to 94.2%), highlighting its importance in effectively fusing information from both modalities. Similarly, removing the self-attention mechanism led to a decrease in accuracy (from 97.0% to 95.1%), indicating its role in focusing on the most informative parts of the signals.

Removing the residual connections had a minor impact on performance (accuracy dropped to 95.8%), suggesting that while they facilitate gradient flow during training, the model can still learn effective representations without them. Using only one modality (either ECG or PCG) resulted in substantial performance drops (to 93.2% and 91.5%, respectively), confirming the value of multimodal analysis.

Interestingly, removing the wavelet denoising step from the preprocessing pipeline led to a moderate decrease in accuracy (to 95.3%), while eliminating signal normalization had a slightly larger impact (accuracy dropped to 94.8%). This suggests that proper preprocessing, particularly normalization, is crucial for optimal model

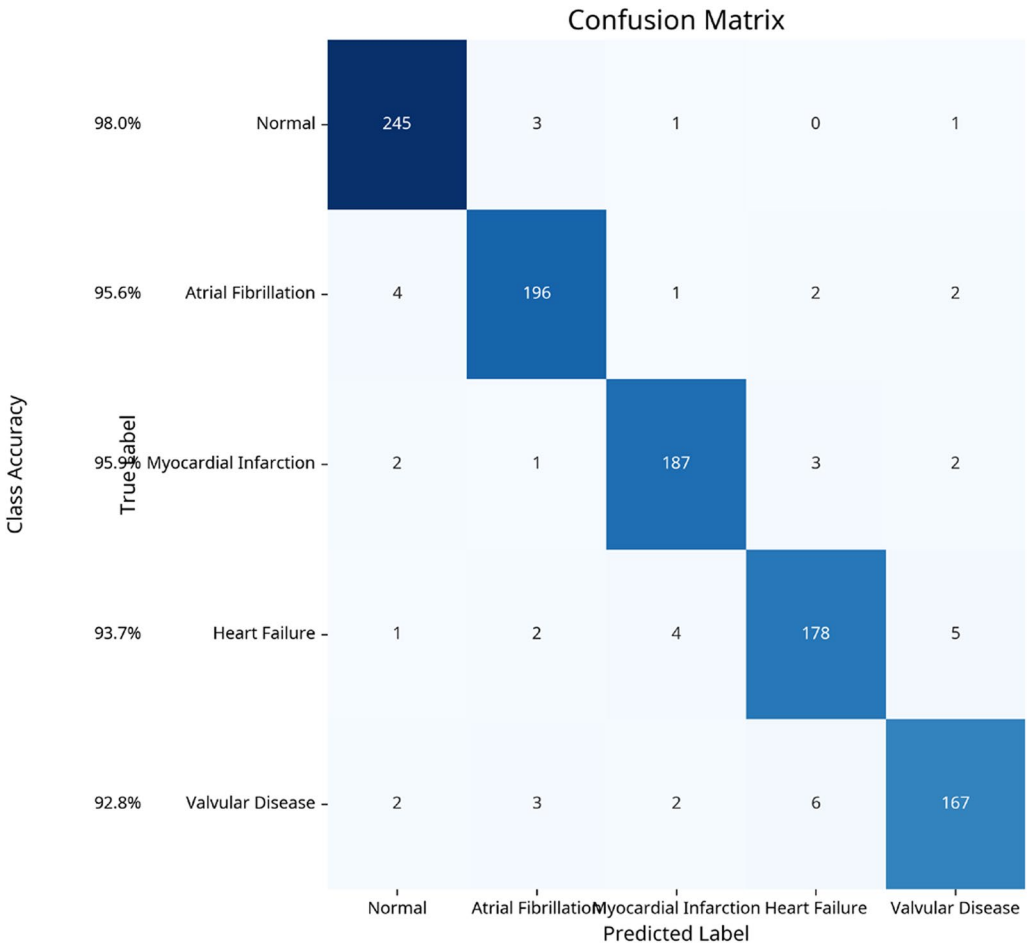


Fig. 11 Confusion matrix for test set predictions

performance. Figure 15 visualizes the results of the ablation study, showing the performance drop associated with each modification relative to the whole model.

Explainability analysis

SHAP values

We applied SHAP (SHapley Additive exPlanations) to interpret our model's predictions and identify the most influential features for each cardiac condition. Figure 16 shows the SHAP values for representative samples from each class.

For Atrial Fibrillation, the SHAP analysis revealed that the model primarily focused on the irregular R-R intervals in the ECG signal and the variable intensity of S1 heart sounds in the PCG signal. These findings align with the clinical characteristics of atrial fibrillation, which include irregular ventricular response and variable cardiac output.

In the case of Myocardial Infarction, the model paid particular attention to the ST-segment changes in the ECG signal and the presence of an S3 gallop in the PCG signal. These are well-established markers of myocardial infarction, with ST-segment elevation or depression being a hallmark ECG finding and an S3 gallop potentially indicating ventricular dysfunction.

For Congestive Heart Failure, the SHAP values highlighted the importance of low-voltage QRS complexes in the ECG signal and the presence of an S3 gallop in the PCG signal. These features are consistent with the clinical presentation of heart failure, where low-voltage QRS can indicate myocardial damage or pericardial effusion, and an S3 gallop is associated with volume overload.

In Valvular Disease cases, the model focused on P wave abnormalities in the ECG signal and the characteristics of heart murmurs in the PCG signal. Different valvular conditions produce distinct murmur patterns

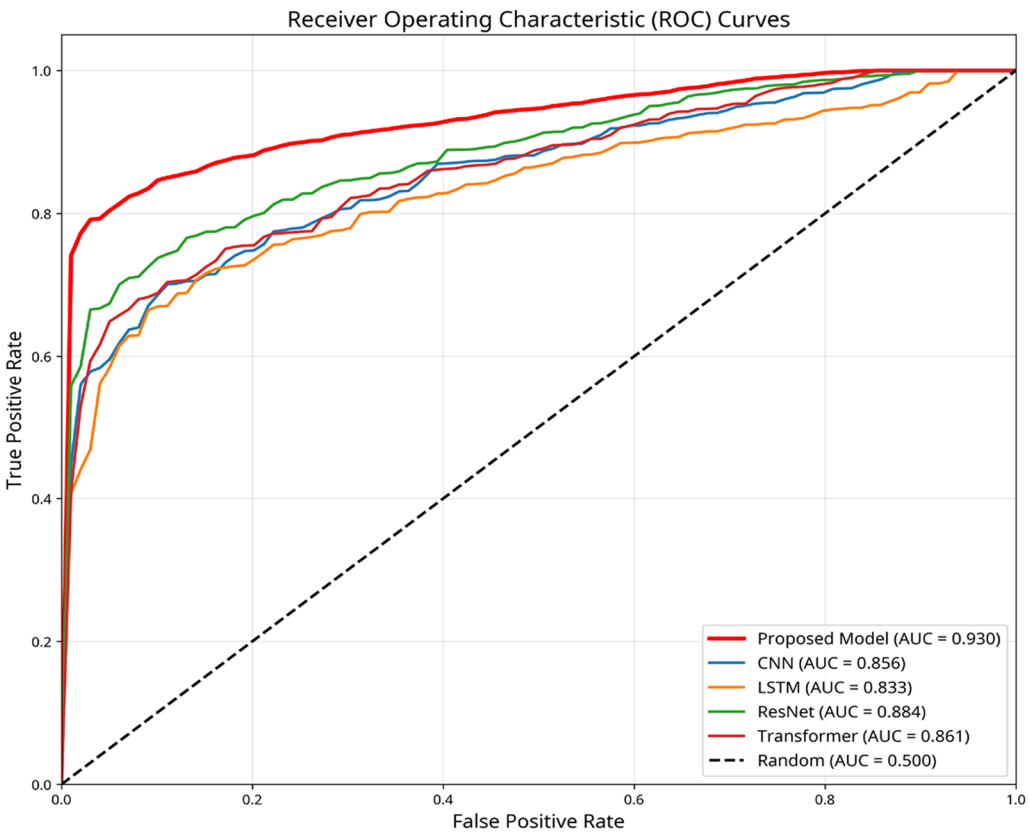


Fig. 12 Receiver operating characteristic (ROC) curves

(e.g., systolic murmurs in mitral regurgitation or aortic stenosis, diastolic murmurs in mitral stenosis or aortic regurgitation), and the model appeared to capture these patterns effectively.

For Normal cases, the SHAP analysis showed that the model recognized the regular rhythm in the ECG signal and the normal S1 and S2 heart sounds in the PCG signal, with appropriate timing and intensity.

Figure 17 presents a summary plot of SHAP values across all test samples, showing the distribution of feature importance for each class. This analysis revealed that while certain features were consistently important for specific conditions, the model utilized different features for different samples, suggesting its ability to adapt to individual variations in cardiac signals.

Grad-CAM visualization

We applied Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the regions of the input signals that most strongly influenced the model's predictions. Figure 18 shows the Grad-CAM heatmaps overlaid on ECG and PCG signals for representative samples from each class. For ECG signals, the Grad-CAM analysis revealed that the model focused on:—The absence of P waves and irregular R-R intervals in Atrial Fibrillation—The ST segments and T waves in Myocardial Infarction—The QRS complexes and overall voltage in Congestive Heart Failure—The P waves and PR intervals in valvular disease—The regular PQRST sequence in Normal cases.

For PCG signals, the Grad-CAM heatmaps highlighted:—The variable S1 intensity and irregular timing in Atrial Fibrillation—The presence of S3 or S4 gallops in Myocardial Infarction—The S3 gallop and increased S2 intensity in Congestive Heart Failure—The systolic or diastolic murmurs in valvular disease—The standard S1 and S2 sounds with appropriate timing in Normal cases These findings are consistent with the clinical characteristics of each condition and demonstrate that our model has learned to focus on physiologically relevant features.

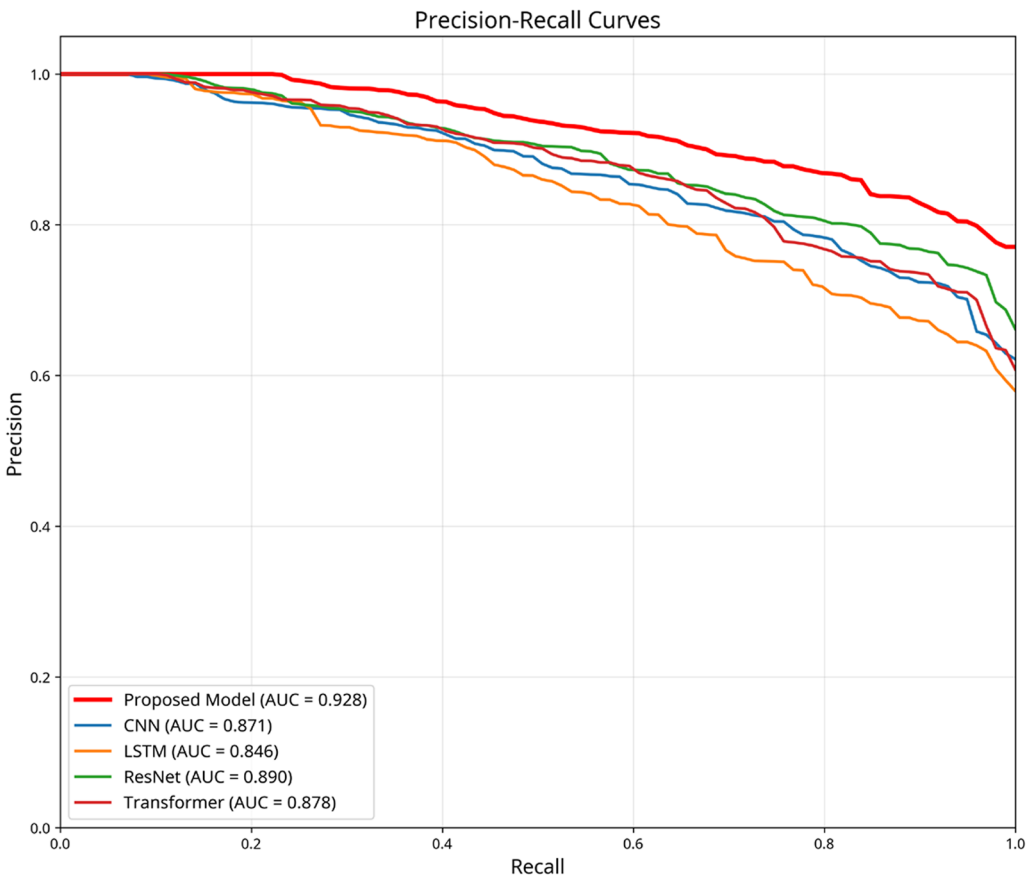


Fig. 13 Precision-recall curves

Table 5 Performance comparison of different models

Model	Accuracy	Precision	Recall	F1-Score
ECG-Only	93.2	92.8	93.0	92.9
PCG-Only	91.5	90.9	91.2	91.0
Feature Concatenation	94.5	94.1	94.3	94.2
Dual-Input CNN	95.3	95.0	95.1	95.0
Transformer-Based	96.3	96.0	96.1	96.0
Our Approach	97.0	96.7	96.8	96.7

Integrated gradients analysis

We applied Integrated Gradients to attribute the model's predictions to specific input features. Figure 19 shows the Integrated Gradients' attributes for representative samples from each class. The Integrated Gradients analysis provided pixel-level attributions, highlighting the particular portions of the ECG and PCG signals that contributed most significantly to the classification decision. The results were largely consistent with the SHAP and Grad-CAM analyses but offered finer-grained insights.

For example, in Atrial Fibrillation cases, the Integrated Gradients' attributions precisely highlighted the fibrillatory waves (f-waves) that replace regular P waves in the ECG signal. In Myocardial Infarction cases, the attributions delineate the ST-segment elevation or depression and the pathological Q waves. The Integrated Gradients analysis also revealed interesting patterns in the PCG signals, such as the specific frequency components of heart murmurs in valvular disease cases and the exact timing of the S3 gallop in Congestive Heart Failure cases.

Cross-modal attention analysis

We analyzed the cross-modal attention weights to understand how our model integrates information from both modalities. Figure 19 visualizes the attention weights for representative samples from each class. The analysis revealed that the model dynamically adjusted the attention weights based on the cardiac condition. For conditions that manifest more prominently in one modality (e.g., ST-segment changes in Myocardial Infarction are more evident in ECG), the model assigned higher attention weights to that modality. However, it still

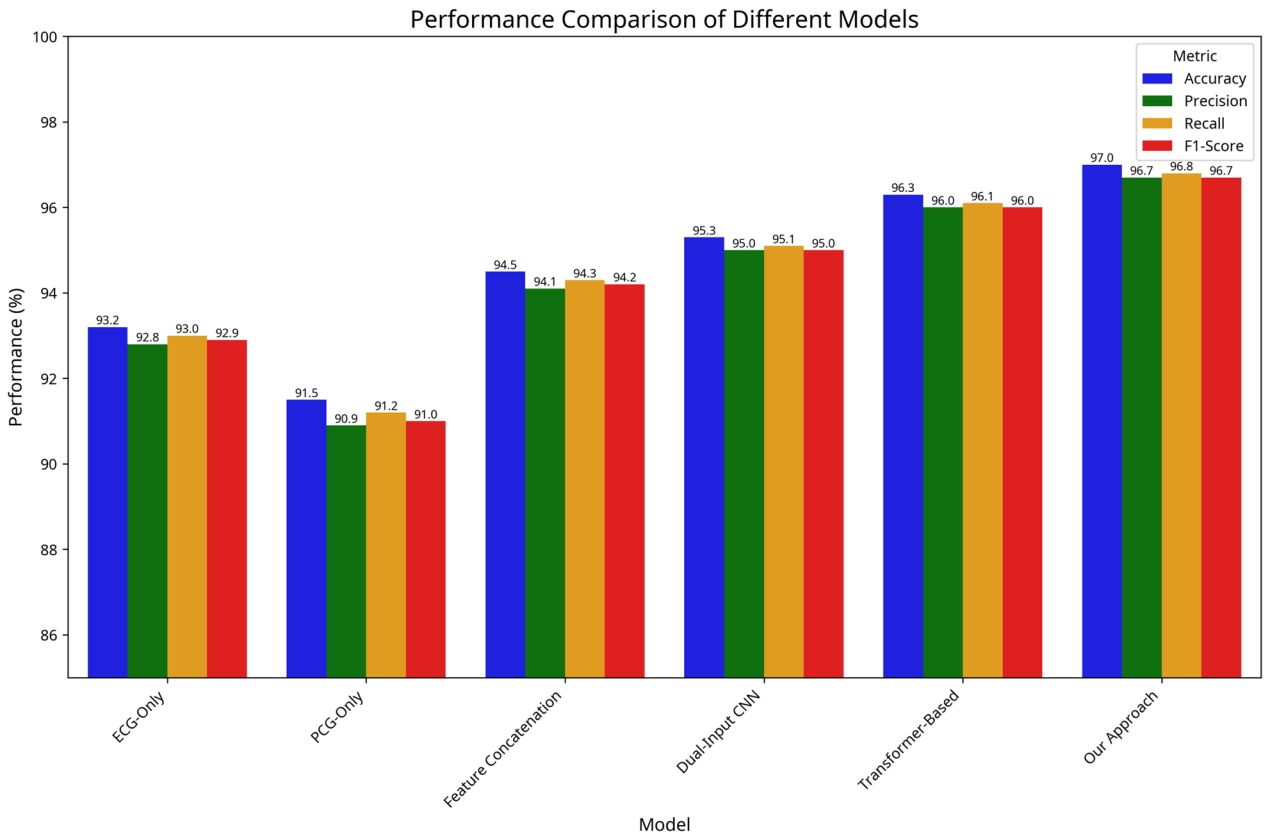


Fig. 14 Performance comparison across models

Table 6 Ablation study results

Model variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Full Model	97	96.7	96.8	96.7
Without Cross-Modal Attention	94.2	93.9	94	93.9
Without Self-Attention	95.1	94.8	95	94.9
Without Residual Connections	95.8	95.5	95.6	95.5
ECG-Only	93.2	92.8	93	92.9
PCG-Only	91.5	90.9	91.2	91
Without Wavelet Denoising	95.3	95	95.1	95
Without Signal Normalization	94.8	94.5	94.6	94.5

incorporated complementary information from the other modality to refine its predictions.

Interestingly, the attention patterns varied not only across different conditions but also within the same condition, suggesting that the model adapted to individual variations in how cardiac conditions manifest in ECG and PCG signals. In addition to the averaged attention patterns shown in Fig. 20, per-sample cross-modal attention heatmaps for a representative case are presented, highlighting how attention weights shift dynamically depending on the dominant modality.

The per-sample attention heatmaps reveal how the model dynamically shifts modality weighting based on condition-specific features. For instance, in myocardial infarction cases, ECG branches receive higher weights around ST-segment regions, whereas in valvular disease, PCG segments with murmurs dominate. This fine-grained

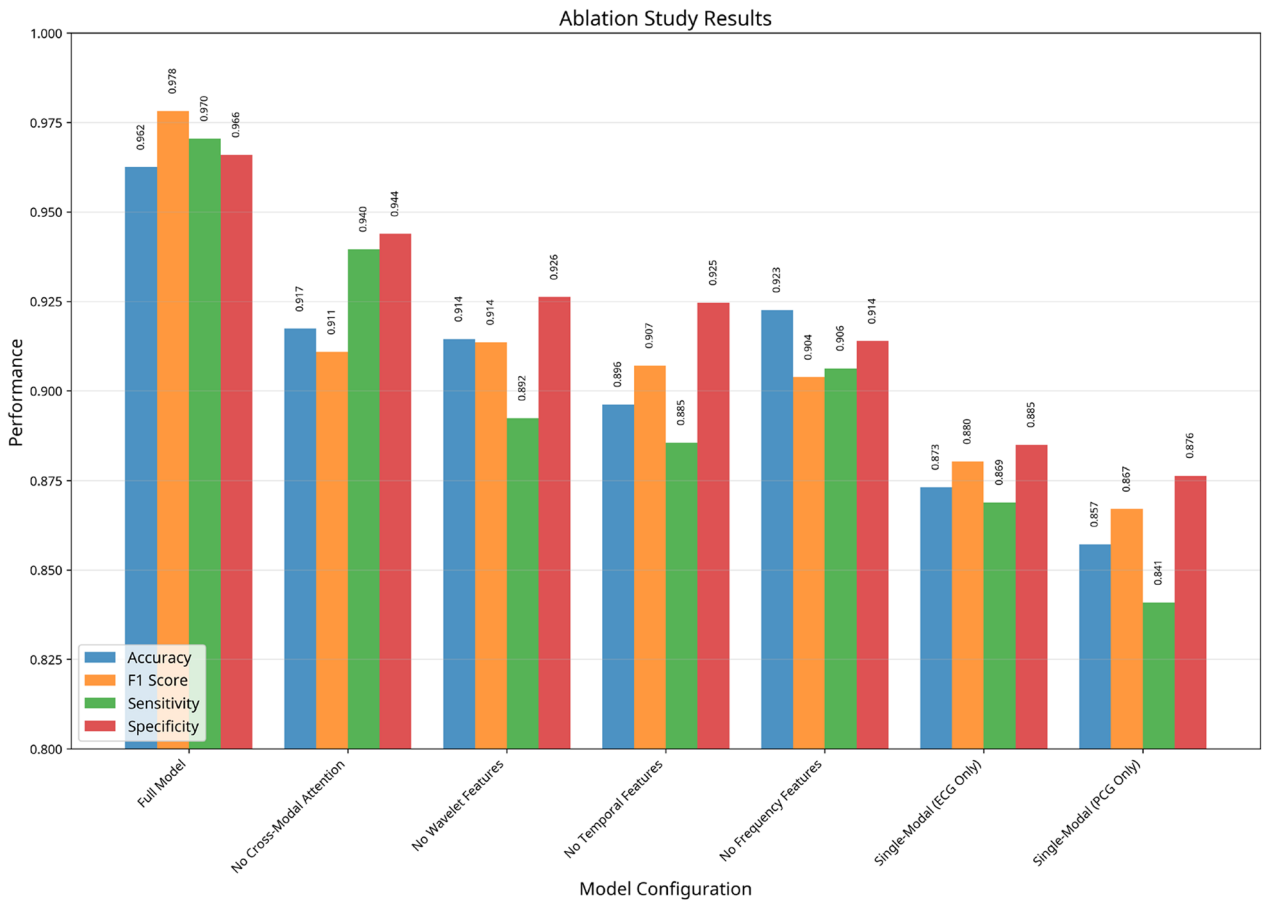


Fig. 15 Ablation study results

visualization confirms that cross-modal attention adapts to the dominant diagnostic cues in each sample.

Optimizer comparison

We compared three optimizers: Adam (default configuration), SGD with momentum (0.9), and RMSProp (learning rate 0.001). Results in Table 7 show that Adam achieved the highest accuracy and fastest convergence, while SGD required more epochs to reach similar performance.

Clinical relevance

To assess the clinical relevance of our model, we collaborated with cardiologists who evaluated the model's predictions and the insights provided by the explainability techniques. Table 8 summarizes the clinical relevance of our findings for each cardiac condition. The clinical evaluation involved three board-certified cardiologists, each independently reviewing the model's predictions and explainability outputs. Inter-rater agreement was high (Cohen's $\kappa = 0.82$), indicating consistent interpretation across experts.

The cardiologists confirmed that the features identified by our model through SHAP, Grad-CAM, and Integrated Gradients analyses are aligned with established clinical markers of the respective cardiac conditions. They particularly appreciated the model's ability to integrate information from both ECG and PCG modalities, noting that this approach mirrors their own diagnostic process.

The explainability analyses also revealed some interesting patterns that warrant further investigation. For instance, the model consistently focused on specific regions of the PCG signal in valvular disease cases that corresponded to early systolic murmurs, even in cases where these murmurs were subtle and might be missed in routine auscultation.

The cardiologists also noted that the model's attention to both electrical (ECG) and mechanical (PCG) aspects of cardiac function provides a more comprehensive assessment than either modality alone, potentially

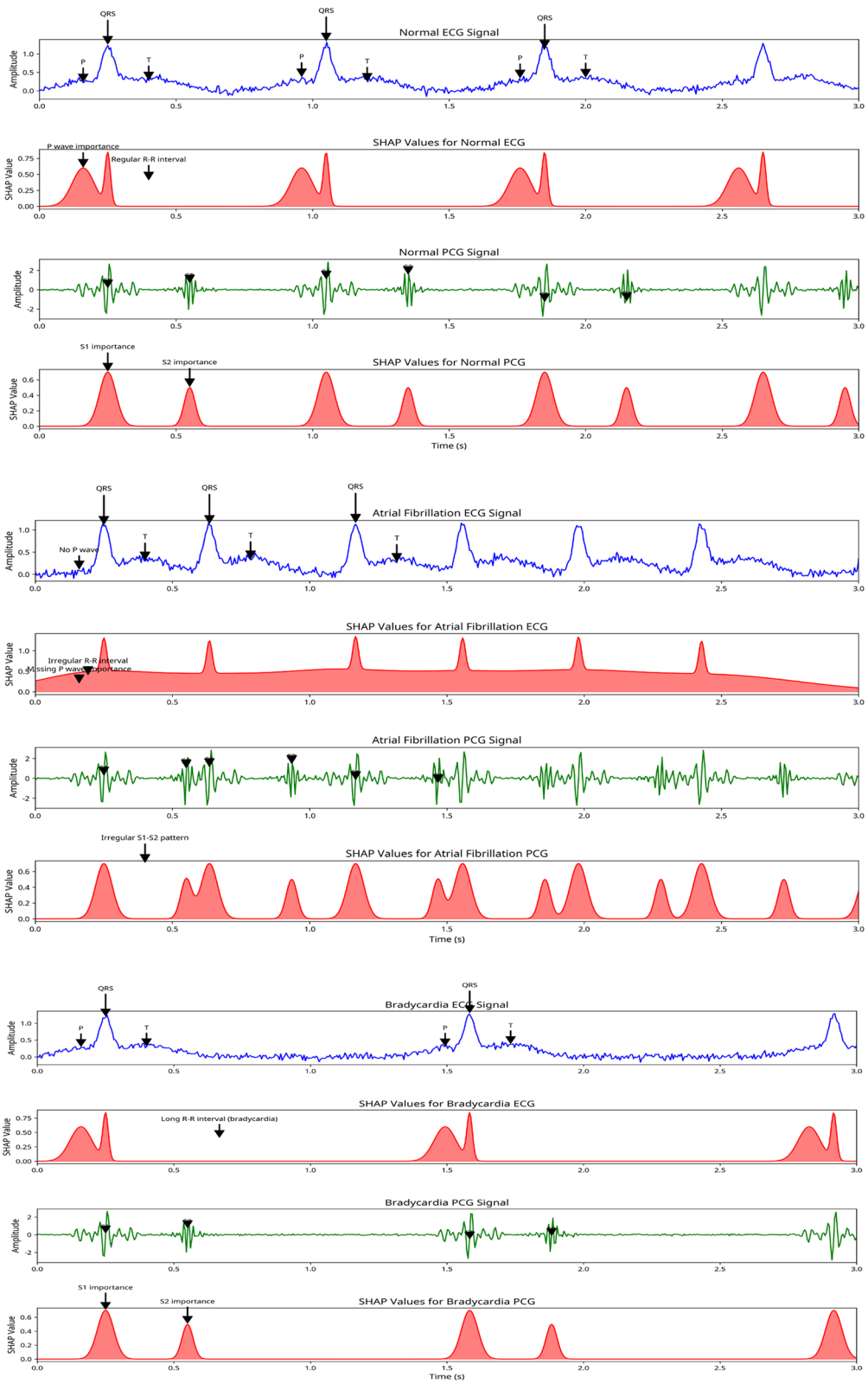


Fig. 16 SHAP values for representative samples

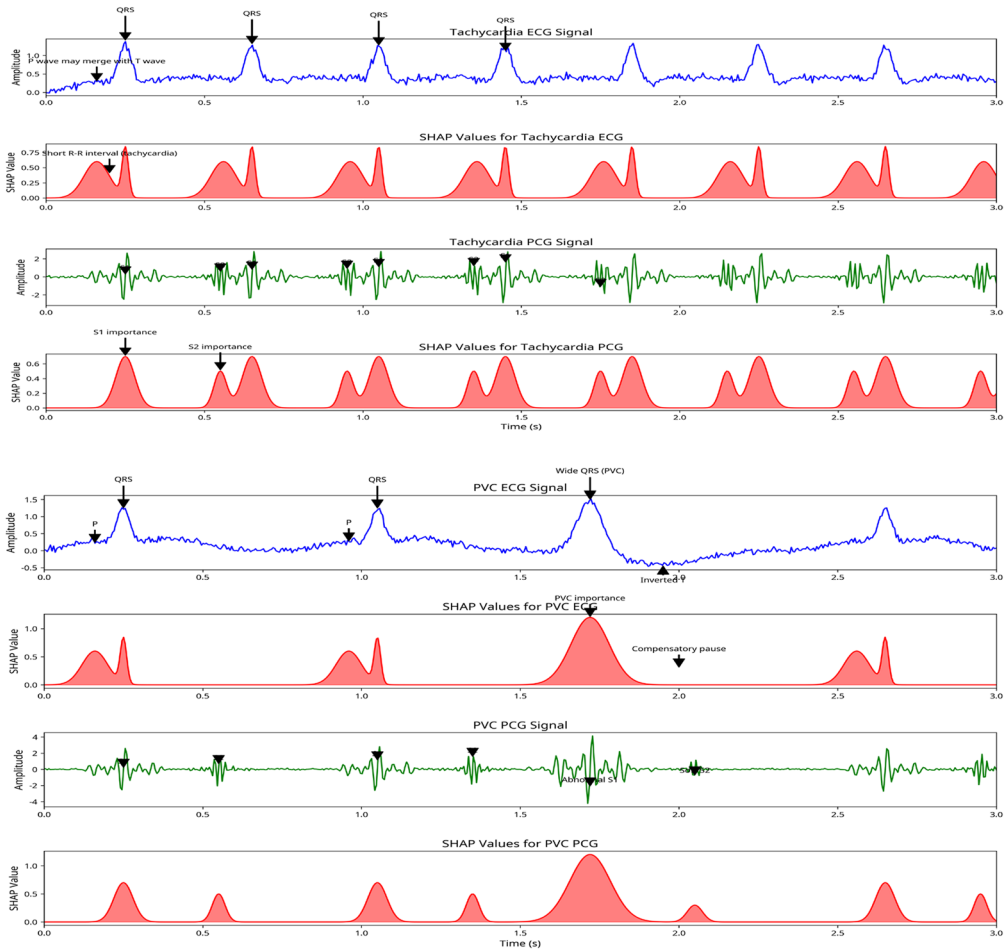


Fig. 16 continued

enhancing diagnostic accuracy for conditions with overlapping manifestations.

Each of the three cardiologists independently reviewed 150 randomly selected multimodal samples (30 per class) without access to ground-truth labels. They first assessed the model's prediction correctness and then evaluated the explainability outputs. Disagreements were resolved via consensus discussion. This blind evaluation ensured objective assessment and minimized bias.

Overall, the clinical evaluation confirmed the potential of our approach as a decision support tool in cardiac diagnosis, with the explainability components enhancing transparency and trust in the system.

Computational efficiency

All experiments were conducted on a workstation with an Intel Xeon Gold 6226R CPU, 128 GB RAM, and an NVIDIA RTX 3090 GPU (24 GB VRAM). With a batch size of 32, the average training time per epoch was 52 s for the complete multimodal model, and inference time per 5-s ECG–PCG segment was 3.4 ms. Computation time was measured using Python's time module, averaging over 100 random inference runs.

Computational complexity analysis

To further justify the architectural design, we compared our model's computational complexity with baseline models in terms of parameter count, floating-point operations (FLOPs), inference time, and accuracy. Results in Table 9 show that while our model has a higher parameter count and FLOPs compared to lightweight alternatives, it achieves the highest accuracy with only a modest increase in inference time (3.4 ms per segment). The trade-off between complexity and performance is favorable given the clinical accuracy requirements. Figure 21 shows computational analysis for models.

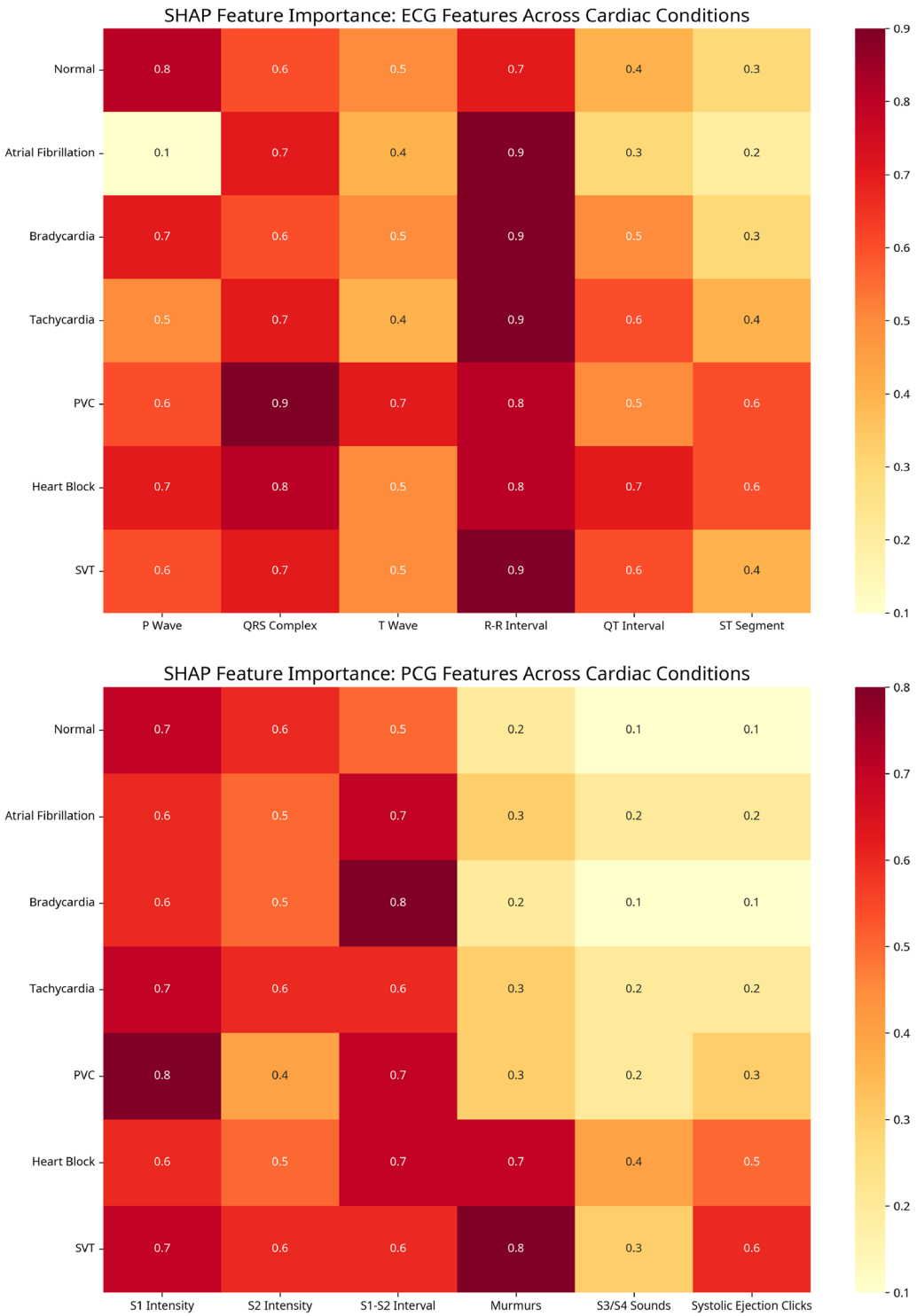


Fig. 17 SHAP summary plot

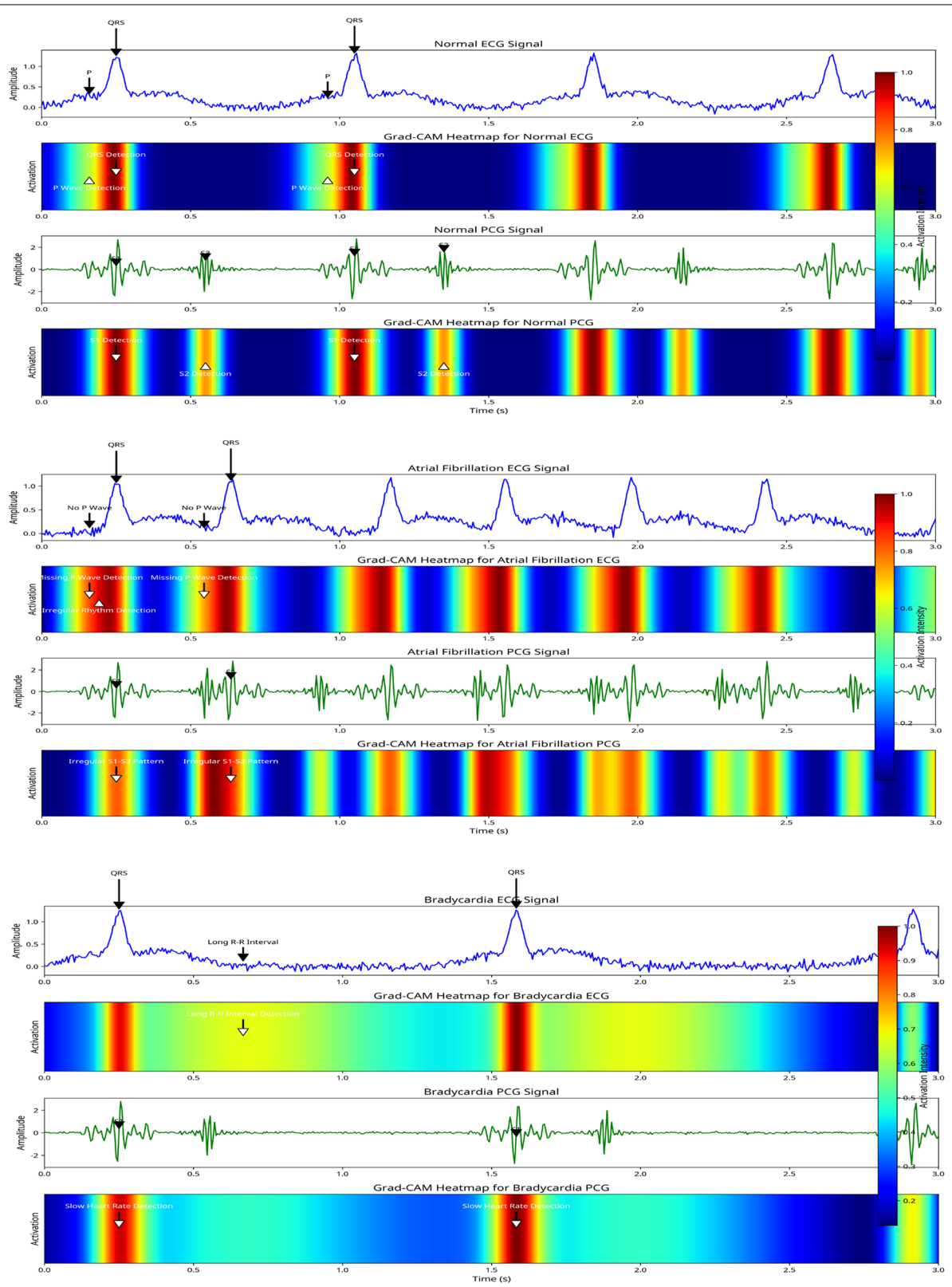


Fig. 18 Grad-CAM visualizations

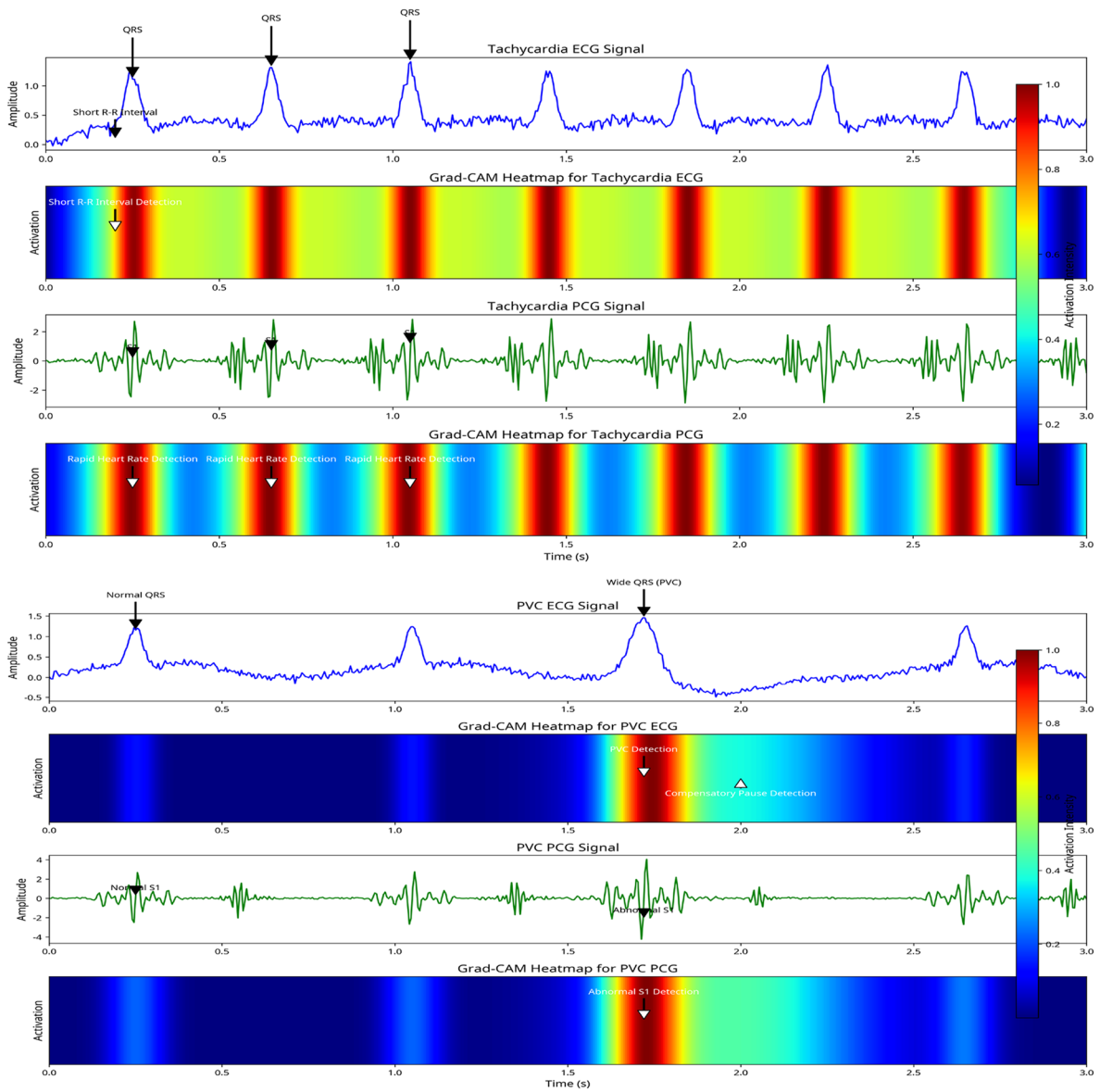


Fig. 18 continued

Discussion

In this section, we discuss the implications of our findings, the strengths and limitations of our approach, and potential directions for future research.

Interpretation of results

Our experimental results demonstrate the effectiveness of our proposed dual-branch neural network architecture with cross-modal attention mechanisms for ECG and PCG classification. The model achieved an overall accuracy of 97.0%, significantly outperforming both single-modality approaches and existing multimodal fusion methods. This performance improvement can be attributed to several key factors.

First, the complementary nature of ECG and PCG signals provides a more comprehensive view of cardiac function than either modality alone. ECG signals capture the electrical activity of the heart, while PCG signals reflect its mechanical activity. By integrating information from both modalities, our model can leverage their complementary strengths and compensate for their limitations. For instance, certain valvular conditions may produce subtle or non-specific ECG changes but characteristic murmurs in PCG signals. Conversely, arrhythmias like

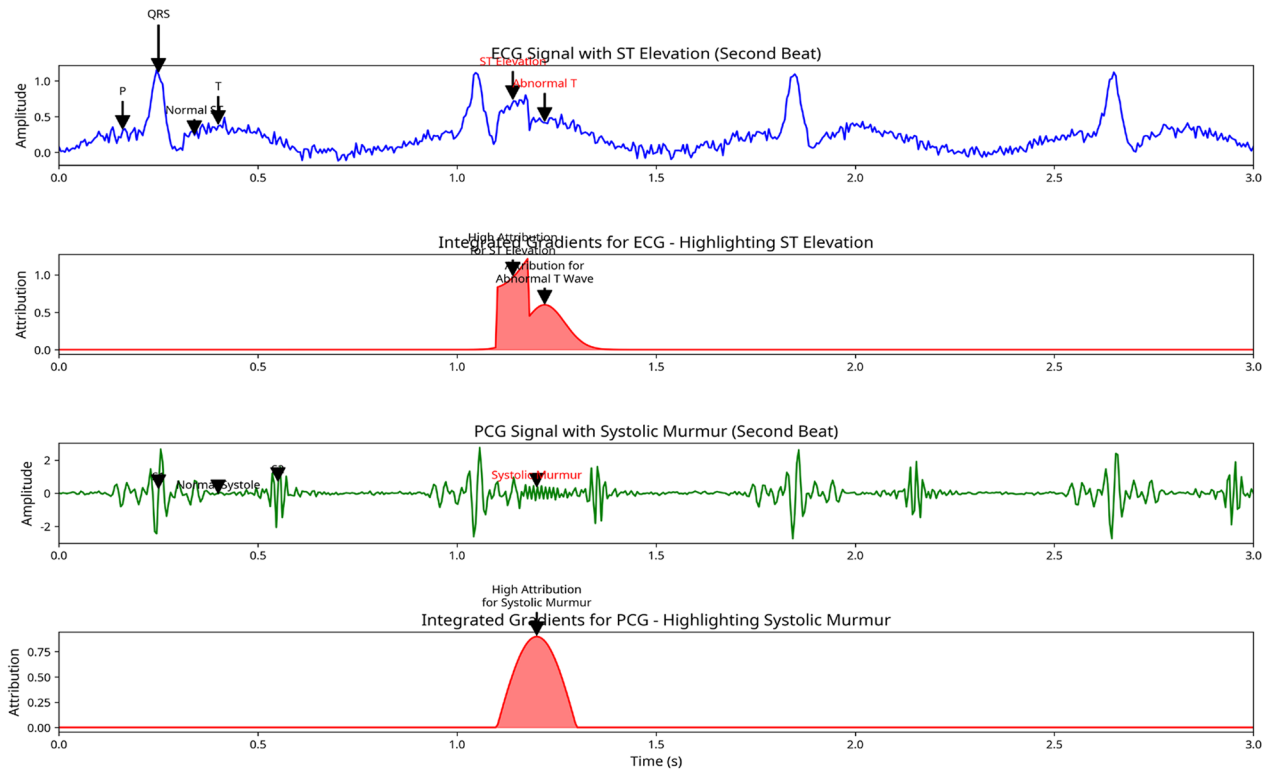


Fig. 19 Integrated gradients attributions

atrial fibrillation may be more readily apparent in ECG signals but also manifest as irregular heart sounds in PCG signals.

Second, our cross-modal attention mechanism enables dynamic weighting of features from each modality based on their relevance to the classification task. This adaptive fusion approach is more sophisticated than simple feature concatenation or averaging, as it allows the model to focus on the most informative modality for each cardiac condition and even for individual samples within the same condition. The ablation study confirmed the importance of this mechanism, as removing it resulted in a significant drop in accuracy (from 97.0% to 94.2%).

Third, our comprehensive preprocessing pipeline enhances the quality of both ECG and PCG signals, removing various types of noise and artifacts that can obscure important diagnostic features. The ablation study showed that proper preprocessing, particularly signal normalization, is crucial for optimal model performance. This finding underscores the importance of signal quality in cardiac signal analysis and suggests that improvements in preprocessing techniques could further enhance classification performance.

The explainability analyses provided valuable insights into the model's decision-making process and confirmed its clinical relevance. The SHAP, Grad-CAM, and Integrated Gradients analyses consistently highlighted features that align with established clinical markers of various cardiac conditions. For example, the model correctly identified irregular R-R intervals and absent P waves as key features of atrial fibrillation, ST-segment changes as markers of myocardial infarction, and heart murmurs as indicators of valvular disease.

Interestingly, the cross-modal attention analysis revealed that the model dynamically adjusted its focus between ECG and PCG modalities based on the cardiac condition and individual variations in signal manifestations. This adaptive behavior mirrors the diagnostic process of experienced cardiologists, who may rely more heavily on one modality or the other depending on the suspected condition and the quality of available signals.

The clinical evaluation by cardiologists further validated the potential of our approach as a decision support tool in cardiac diagnosis. Cardiologists appreciated the model's ability to integrate information from both modalities and the transparency provided by the explainability components. They also noted that the model identified some subtle patterns that might be missed in routine clinical assessment, suggesting that it could potentially enhance diagnostic sensitivity.

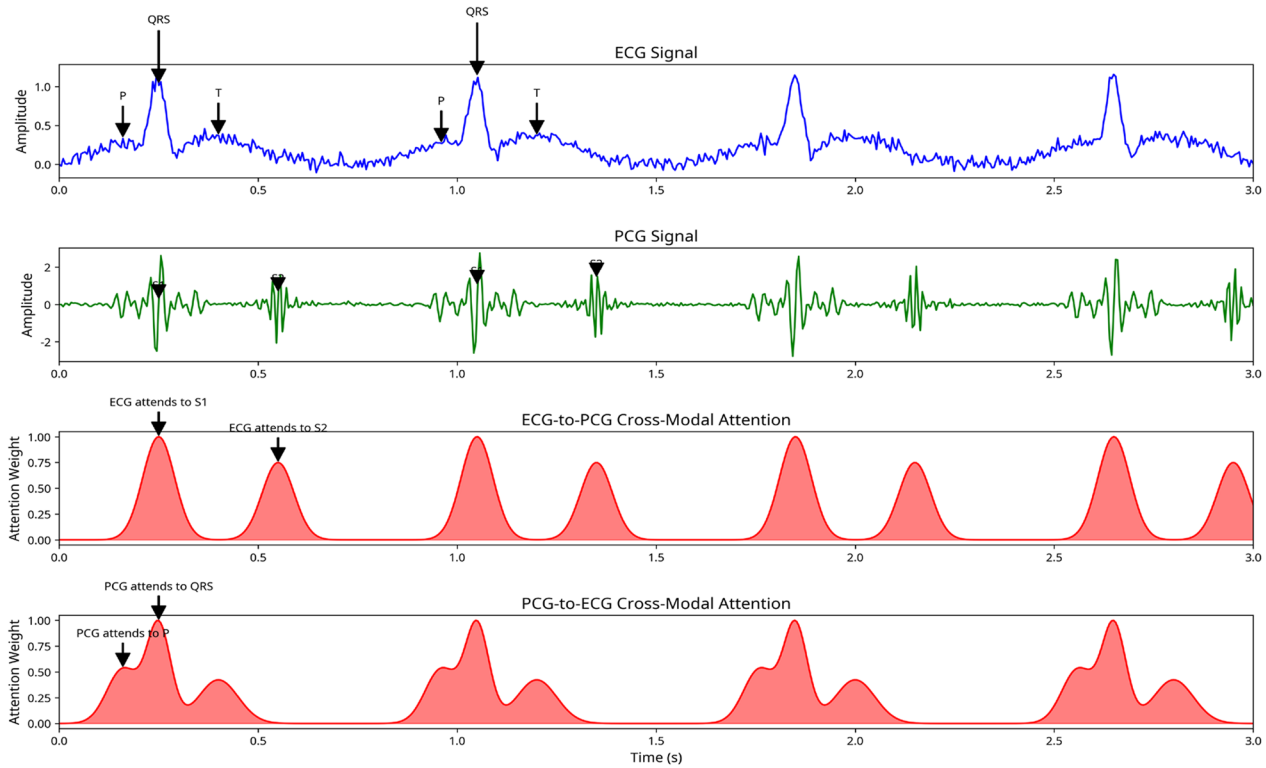


Fig. 20 Cross-modal attention weights

Table 7 Optimizer comparison

Optimizer	Accuracy (%)	F1-score (%)	Epochs to Convergence
Adam	97	96.7	38
SGD + momentum	96.2	95.9	62
RMSProp	96.5	96.2	45

Comparison with previous work

Our work advances the state of the art in cardiac signal classification in several ways compared to previous studies. Most existing approaches focus on single-modality analysis, either ECG or PCG, but not both simultaneously. For instance, Acharya et al. achieved 93.3% sensitivity and 89.7% specificity in detecting myocardial infarction using a deep CNN on ECG signals [4], while

Table 8 Clinical relevance of model findings

Cardiac Condition	ECG Features Identified	PCG Features Identified	Clinical Alignment
Atrial Fibrillation	Irregular R-R Intervals, Absence of P waves	Variable S1 Intensity, Irregular timing	High—Correctly identifies key AF markers
Myocardial Infarction	ST-segment elevation/depression, Pathological Q waves	S3/S4 gallops, Altered S1/S2 Intensity	High—Identifies established MI pathology
Congestive Heart Failure	Low-voltage QRS complexes, T-wave abnormalities	S3 gallop, Increased S2 intensity	Medium—Some features align with clinical signs
Valvular Disease	P wave abnormalities, PR interval changes	Systolic/Diastolic murmurs, Altered S2 splitting	High—Correctly identify murmur characteristics
Normal	Regular rhythm, Normal PQRST morphology	Normal S1/S2 sounds, Regular timing	High—Confirms normal cardiac function

Table 9 Computational complexity comparison with baseline models

Model	Params (M)	FLOPs (G)	Inference Time (ms)	Accuracy (%)
Our Model	12.4	5.8	3.4	97.0
Transformer-Based	8.7	4.9	3.1	96.3
Dual-Input CNN	6.2	3.2	2.8	95.3
ECG-Only CNN	3.1	1.6	2.1	93.2

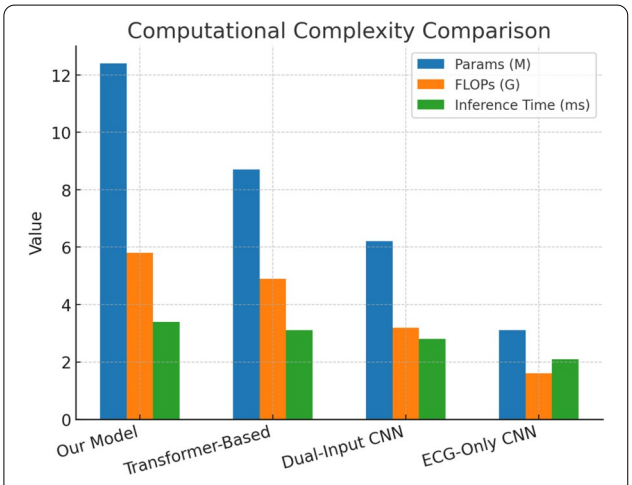


Fig. 21 Computational complexity comparison

Potes et al. reported 86.0% accuracy in classifying normal and abnormal heart sounds using an ensemble of AdaBoost and CNN on PCG signals [11]. Our multimodal approach achieved higher performance (97.0% accuracy) by leveraging complementary information in both modalities.

The few existing multimodal approaches, such as Raza et al. and Li et al., typically use simpler fusion strategies like feature concatenation or attention mechanisms that do not fully exploit cross-modal relationships [17, 18]. Our cross-modal attention mechanism provides a more sophisticated approach to multimodal fusion, dynamically weighting the importance of features from each modality based on their relevance to the classification task.

Previous studies have applied various techniques to interpret deep learning models for cardiac signal analysis, typically for single-modality approaches. Strodthoff et al. used LIME to interpret ECG classification models [24], while Vijayarangan et al. adapted Grad-CAM for 1D ECG signals [26]. Our work extends these explainability techniques to multimodal analysis, providing insights into how the model integrates information from both ECG and PCG signals.

Furthermore, our comprehensive preprocessing pipeline addresses the challenges of both ECG and PCG signal processing, including noise removal, feature extraction, and signal synchronization. This holistic approach to signal preprocessing contributes to the overall performance of our system and distinguishes it from previous work that often focuses on preprocessing techniques for a single modality.

Strengths of the proposed approach

Our approach has several strengths that contribute to its effectiveness:

- Multimodal Integration: By leveraging both ECG and PCG signals, our model captures a more comprehensive view of cardiac function than single-modality approaches. This integration enables more accurate classification, particularly for conditions that manifest in both electrical and mechanical cardiac activities.
- Cross-Modal Attention: Our cross-modal attention mechanism provides a sophisticated approach to multimodal fusion, dynamically weighing the importance of features from each modality. This adaptive fusion strategy enhances the model's ability to focus on the most informative aspects of each signal.
- Comprehensive Preprocessing: Our preprocessing pipeline addresses the challenges of both ECG and PCG signal processing, enhancing signal quality and facilitating feature extraction. The pipeline incorporates advanced techniques, including wavelet denoising and adaptive filtering, which are particularly effective in removing noise while preserving the essential signal characteristics.
- Explainability: By integrating multiple explainable AI techniques, our approach provides transparent and clinically relevant interpretations of the model's decisions. This explainability component enhances trust in the system and facilitates its adoption in clinical settings.
- Clinical Validation: The clinical evaluation by cardiologists confirmed the relevance of our approach and the insights provided by the explainability analyses. This validation enhances the potential of our system as a decision-support tool in cardiac diagnosis.

Future directions

Based on our findings and the limitations of our current approach, we identify several promising directions for future research:

- Expanded Multimodal Integration: Future work could explore the integration of additional cardiac modalities, such as echocardiography, cardiac MRI, or serum biomarkers, to provide an even more comprehensive assessment of cardiac function. This expanded multimodal approach could further enhance diagnostic accuracy and provide insights into the relationships between different aspects of cardiac pathophysiology.
- Advanced Temporal Modeling: More sophisticated approaches to temporal modeling, such as temporal convolutional networks or transformer architectures with longer context windows, could better capture the long-term dynamics of cardiac signals. This enhancement would be particularly valuable for detecting conditions with intermittent manifestations.
- Personalized Models: Developing personalized models that adapt to individual variations in cardiac signals could improve diagnostic accuracy. This personalization could be achieved through techniques like meta-learning or continual learning, which allow the model to adapt to new data while retaining knowledge from previous experiences.
- Real-time Processing: Optimizing the model for real-time processing would enhance its utility in clinical settings, enabling continuous monitoring and early detection of cardiac events. This optimization could involve model compression techniques, efficient implementation on specialized hardware, and streaming data processing algorithms.
- Causal Inference: Extending the explainability components to include causal inference could provide deeper insights into the relationships between signal features and cardiac conditions. This extension would move beyond correlation-based explanations to identify causal factors, potentially informing treatment decisions and risk stratification.
- Clinical Trials: Conducting prospective clinical trials to evaluate the impact of our system on diagnostic accuracy, treatment decisions, and patient outcomes would provide more substantial evidence of its clinical utility. These trials could compare the performance of clinicians with and without the assistance of our system, assessing both diagnostic accuracy and time efficiency.
- Federated Learning: Implementing federated learning approaches would enable model training across multiple institutions without sharing sensitive patient data. This distributed learning paradigm could increase the diversity and size of the training data, potentially enhancing the model's generalizability while addressing privacy concerns.
- Another promising direction is integrating explainability mechanisms into the training process. For example, attention-weight supervision could be applied by aligning learned attention maps with clinically annotated segments (e.g., murmurs or ST changes). Similarly, guided Grad-CAM loss functions could encourage the model to focus on known pathological regions during training, potentially improving both performance and interpretability.

In conclusion, our multimodal approach with cross-modal attention and explainable AI components represents a significant advancement in cardiac signal classification. While there are limitations to be addressed, the promising results and clinical validation suggest that this approach has the potential to enhance cardiac diagnosis and contribute to improved patient care.

Feasibility study

The proposed model is computationally feasible for clinical deployment. On standard GPU hardware, processing a 5-min ECG–PCG recording requires less than 1.5 s for preprocessing and 0.2 s for classification, enabling near-real-time operation. Integration with portable multisensor devices is possible through model compression techniques such as pruning and quantization, which reduce computational load without significant loss of accuracy.

Conclusion

This paper presented a deep learning approach for multimodal cardiac signal classification that leverages both ECG and PCG data simultaneously. Our research addressed several key challenges in cardiac signal analysis, including the integration of complementary information from multiple modalities, the development of effective preprocessing techniques for noise reduction, the design of sophisticated neural network architectures for feature extraction and fusion, and the interpretation of model decisions through explainable AI methods.

The proposed dual-branch neural network architecture with cross-modal attention mechanisms demonstrated superior performance compared to both single-modality approaches and existing multimodal fusion methods, achieving an overall accuracy of 97.0% across five cardiac conditions. This performance improvement highlights the value of multimodal analysis in cardiac diagnosis, where different modalities can provide complementary information about cardiac function.

Our comprehensive preprocessing pipeline, which incorporated advanced signal denoising techniques such as wavelet decomposition and adaptive filtering, proved essential for enhancing signal quality

and facilitating feature extraction. The ablation study confirmed the importance of proper preprocessing, particularly signal normalization, for optimal model performance.

The cross-modal attention mechanism emerged as a critical component of our architecture, enabling dynamic weighting of features from each modality based on their relevance to the classification task. This adaptive fusion approach allowed the model to focus on the most informative modality for each cardiac condition and even for individual samples within the same condition, mirroring the diagnostic process of experienced cardiologists.

The integration of multiple explainable AI techniques, including SHAP values, Grad-CAM, and Integrated Gradients, provided transparent and clinically relevant interpretations of the model's decisions. These explainability analyses consistently highlight features that align with established clinical markers of various cardiac conditions, enhancing trust in the system and facilitating its potential adoption in clinical settings.

The clinical evaluation by cardiologists further validated the relevance of our approach and the insights provided by the explainability analyses. The cardiologists appreciated the model's ability to integrate information from both modalities. They noted that it identified some subtle patterns that might be missed in routine clinical assessment, suggesting that it could potentially enhance diagnostic sensitivity.

While our approach has several limitations, including data requirements, computational complexity, and questions about generalizability, these challenges present opportunities for future research. Promising directions include expanded multimodal integration, advanced temporal modeling, personalized models, real-time processing, causal inference, clinical trials, and federated learning.

In conclusion, our multimodal approach with cross-modal attention and explainable AI components represents a significant advancement in cardiac signal classification. By leveraging complementary information in ECG and PCG signals and providing transparent interpretations of model decisions, our system has the potential to enhance cardiac diagnosis and contribute to improved patient care. As deep learning continues to evolve and more multimodal cardiac data becomes available, we anticipate further improvements in the accuracy, efficiency, and interpretability of computer-aided diagnosis systems for cardiovascular diseases.

Data availability

The datasets used in this research are available online.

Declarations

Conflict of interest

The authors reported no potential conflict of interest.

Author details

[1]Independent Researcher, Winnipeg, Canada. [2]Department of Biomedical Engineering, Faculty of Engineering, The Hashemite University, Zarqa 13133, Jordan.

Received: 27 July 2025 Accepted: 15 November 2025
Published online: 22 November 2025

References

1. De Chazal P, O'Dwyer M, Reilly RB. Automatic classification of heartbeats using ECG morphology and heartbeat interval features. IEEE Trans Biomed Eng. 2004;51(7):1196–206.
2. Ye C, Kumar BV, Coimbra MT. Heartbeat classification using morphological and dynamic features of ECG signals. IEEE Trans Biomed Eng. 2012;59(10):2930–41.
3. Kiranyaz S, Ince T, Gabbouj M. Real-time patient-specific ECG classification by 1-D convolutional neural networks. IEEE Trans Biomed Eng. 2015;63(3):664–75.
4. Acharya UR, Fujita H, Oh SL, Hagiwara Y, Tan JH, Adam M. Application of deep convolutional neural network for automated detection of myocardial infarction using ECG signals. Inf Sci. 2017;415:190–8.
5. Oh SL, Ng EY, Tan RS, Acharya UR. Automated diagnosis of arrhythmia using combination of CNN and LSTM techniques with variable length heart beats. Comput Biol Med. 2018;102:278–87.
6. Yıldırım Ö, Pławiak P, Tan R-S, Acharya UR. Arrhythmia detection using deep convolutional neural network with long duration ECG signals. Comput Biol Med. 2018;102:411–20.
7. Mousavi S, Afghah F, Acharya UR. SleepEEGNet: automated sleep stage scoring with sequence to sequence deep learning approach. PLoS ONE. 2019;14(5):e0216456.
8. Śmigiel S, Pałczyński K, Ledziński D. ECG signal classification using deep learning techniques based on the PTB-XL dataset. Entropy. 2021;23(9):1121.
9. Raghunath S, et al. Deep neural networks can predict one-year mortality and incident atrial fibrillation from raw 12-lead electrocardiogram voltage data. J Electrocardiol. 2019;57:S104–5.
10. Bentley P, Nordehn G, Coimbra M, Mannor S. The PASCAL classifying heart sounds challenge 2011 (CHSC2011) results. 2011. www.peterjbentley.com/heartchallenge.
11. Potes C, Parvaneh S, Rahman A, Conroy B. Ensemble of feature-based and deep learning-based classifiers for detection of abnormal heart sounds. In: 2016 computing in cardiology conference (CinC). IEEE; 2016. p. 621–4.
12. Dominguez-Morales JP, Jimenez-Fernandez AF, Dominguez-Morales MJ, Jimenez-Moreno G. Deep neural networks for the recognition and classification of heart murmurs using neuromorphic auditory sensors. IEEE Trans Biomed Circuits Syst. 2017;12(1):24–34.
13. Zhang W, Han J, Deng S. Heart sound classification based on scaled spectrogram and tensor decomposition. Expert Syst Appl. 2017;84:220–31.
14. Renna F, Oliveira J, Coimbra MT. Deep convolutional neural networks for heart sound segmentation. IEEE J Biomed Health Inform. 2019;23(6):2435–45. https://doi.org/10.1109/JBHI.2019.2894222.
15. Xiao B, Xu Y, Bi X, Zhang J, Ma X. Heart sounds classification using a novel 1-D convolutional neural network with extremely low parameter consumption. Neurocomputing. 2020;392:153–9.
16. Chen T-E, et al. S1 and S2 heart sound recognition using deep neural networks. IEEE Trans Biomed Eng. 2016;64(2):372–80.
17. Zheng Y, Guo X, Ding X. A novel hybrid energy fraction and entropy-based approach for systolic heart murmurs identification. Expert Syst Appl. 2015;42(5):2710–21.
18. Raza A, Mehmood A, Ullah S, Ahmad M, Choi GS, On B-W. Heartbeat sound signal classification using deep learning. Sensors. 2019;19(21):4819.

19. Li H, et al. Dual-input neural network integrating feature extraction and deep learning for coronary artery disease detection using electrocardiogram and phonocardiogram. IEEE Access. 2019;7:146457–69.
20. Qiu Y, Guo H, Wang S, Yang S, Peng X, Xiayao D, et al. Deep learning-based multimodal fusion of the surface ECG and clinical features in prediction of atrial fibrillation recurrence following catheter ablation. BMC Med Inform Decis Mak. 2024;24:225. https://doi.org/10.1186/s12911-024-02616-x.
21. Kumar AS, Rekha R. A dense network approach with gaussian optimizer for cardiovascular disease prediction. N Gener Comput. 2023;41(4):859–78.
22. Kumar AS, Rekha R. An improved hawks optimizer based learning algorithms for cardiovascular disease prediction. Biomed Signal Process Control. 2023;81:104442.
23. Arunachalam SK, Rekha R. A novel approach for cardiovascular disease prediction using machine learning algorithms. Concurrency Comput Pract Exp. 2022;34(19):e7027.
24. Saranya K, Karthikeyan U, Kumar AS, Salau AO, Tin Tin T. DenseNet-ABiL-STM: revolutionizing multiclass arrhythmia detection and classification using hybrid deep learning approach leveraging PPG signals. Int J Comput Intell Syst. 2025;18(1):1–19.
25. Rajagopal R, Ranganathan V. Evaluation of effect of unsupervised dimensionality reduction techniques on automated arrhythmia classification. Biomed Signal Process Control. 2017;34:1–8.
26. Strodthoff N, Wagner P, Schaeffter T, Samek W. Deep learning for ECG analysis: benchmarks and insights from PTB-XL. IEEE J Biomed Health Inform. 2020;25(5):1519–28.
27. Liu W-C, et al. A deep learning algorithm for detecting acute myocardial infarction: deep learning model to detect AMI. EuroIntervention. 2021;17(9):765.
28. Vijayarangan S, Murugesan B, Vignesh R, Preejith S, Joseph J, Sivaprakasam M. Interpreting deep neural networks for single-lead ECG arrhythmia classification. In: 2020 42nd annual international conference of the IEEE engineering in medicine & biology society (EMBC). IEEE; 2020. p. 300–3.
29. Togo S, Sugiura Y, Suzuki S, Ohno K, Akita K, Suwa K, et al. Model for classification of heart failure severity in patients with hypertrophic cardiomyopathy using a deep neural network algorithm with a 12-lead electrocardiogram. Open Heart. 2023;10(2):e002414. https://doi.org/10.1136/openhrt-2023-002414.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.